AWS Solution Architect Training with
AWS Cloud Practitioner Global Certification

Capstone Projects

Trainer: Aravindraj.G- Nminds Academy

AWS Academy Certified Educator

1.Create an EC2 Windows Instance in AWS

Deepika.D

CSE-Artificial Intelligence & Machine Learning,

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Coimbatore.

Objective

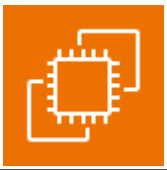
EC2 (Elastic Compute Cloud) is a core service provided by Amazon Web Services (AWS) that enables users to rent virtual servers in the cloud to run applications, store data, or handle computational workloads.

An EC2 Windows instance is a virtual machine running the Microsoft Windows operating system on Amazon Web Services (AWS) EC2 (Elastic Compute Cloud). This allows you to run Windows-based applications, manage Windows Server configurations, and use various Microsoft software (like SQL Server, IIS, etc.) in the cloud without having to manage physical hardware.

Use Cases for EC2 Windows Instances:

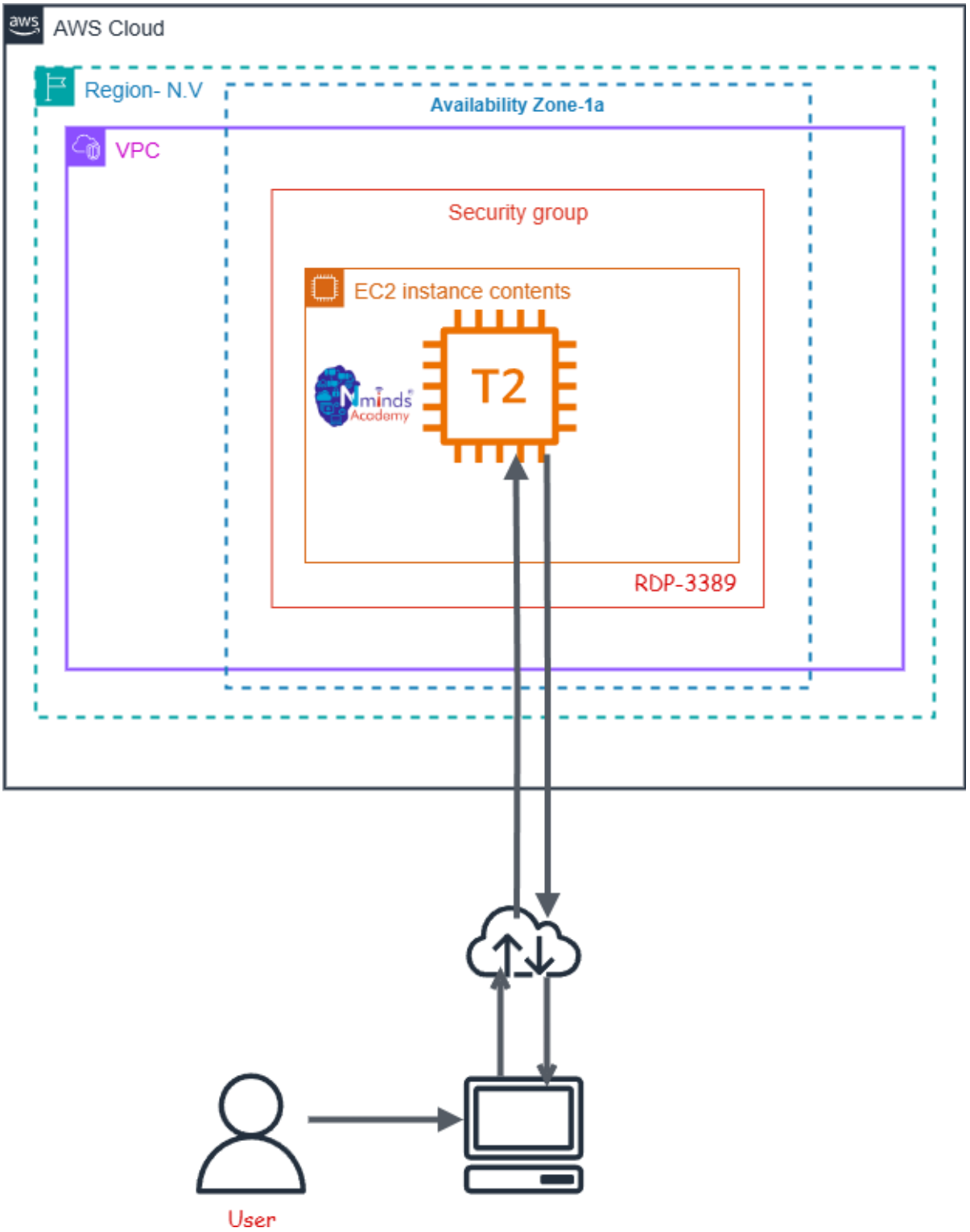
- Web and Application Hosting: Run IIS web servers or other Windows-based web applications in the cloud.
- Database Servers: Host SQL Server databases with EC2 Windows instances, using RDS or EC2 for more control.
- Remote Desktop Applications: Set up RDS (Remote Desktop Services) for remote access to Windows desktops in the cloud.
- Development and Testing: Create development environments with Windows OS to test .NET or other Windows-based applications.

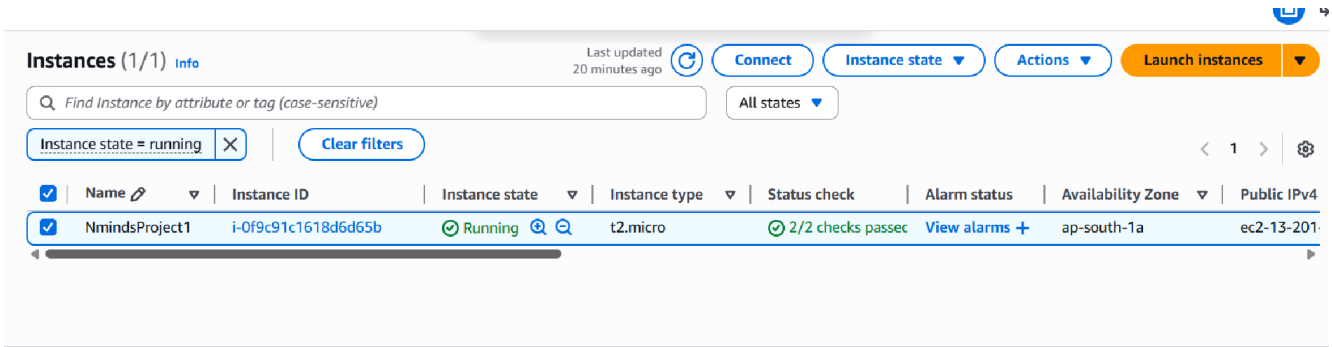
Used Services in AWS:

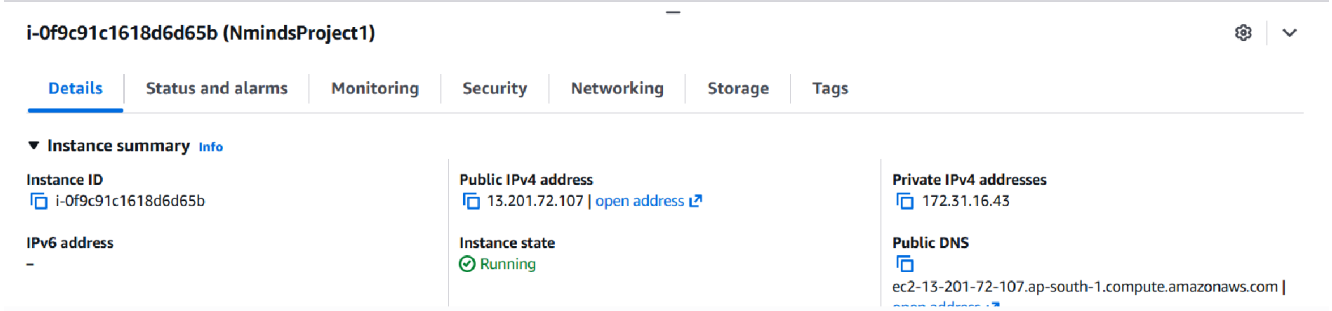


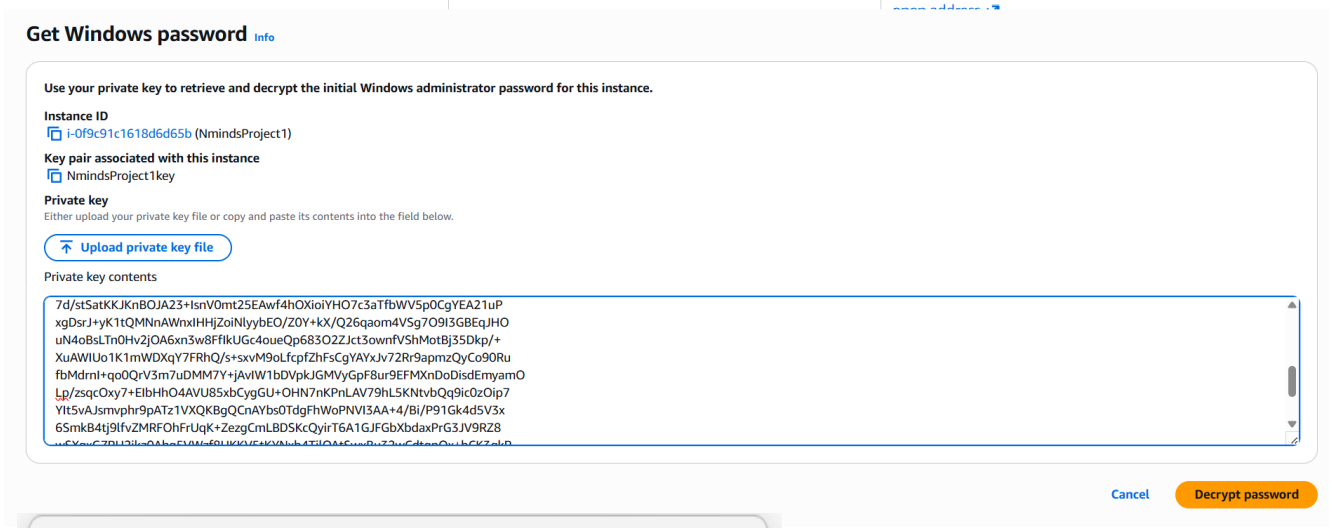
Amazon EC2

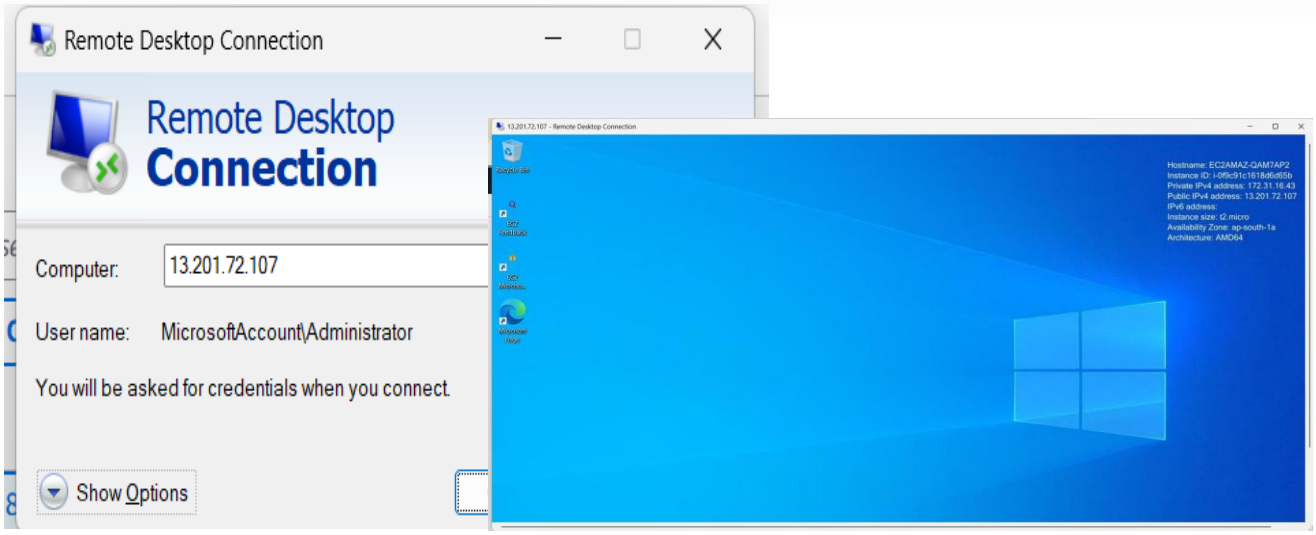
Topology


AWS Cloud
Region- N.V
Availability Zone-1a
VPC
Security group
EC2 instance contents
T2
RDP-3389
User


Instances (1/1) Info
Last updated 20 minutes ago
Connect
Instance state
Actions
Launch instances
Find instance by attribute or tag (case-sensitive)
All states
Instance state = running
Clear filters
Name
Instance ID
Instance state
Instance type
Status check
Alarm status
Availability Zone
Public IPv4
NmindsProject1
i-0f9c91c1618d6d65b
Running
t2.micro
2/2 checks passed
View alarms +
ap-south-1a
ec2-13-201


i-0f9c91c1618d6d65b (NmindsProject1)
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags
Instance summary Info
Instance ID
i-0f9c91c1618d6d65b
Public IPv4 address
13.201.72.107 | open address
Private IPv4 addresses
172.31.16.43
IPv6 address
-
Instance state
Running
Public DNS
ec2-13-201-72-107.ap-south-1.compute.amazonaws.com |


Get Windows password Info
Use your private key to retrieve and decrypt the initial Windows administrator password for this instance.
Instance ID
i-0f9c91c1618d6d65b (NmindsProject1)
Key pair associated with this instance
NmindsProject1key
Private key
Either upload your private key file or copy and paste its contents into the field below.
Upload private key file
Private key contents
7d/stSatKKJKnBOJA23+IsnV0mt25EAwf4hOXioiYHO7c3aTfbWV5p0CgYEA21uP
xgDsrJ+yK1tQMNnAWnxIHHjZoiNIyybEO/Z0Y+kX/Q26qaom4V5g7O9I3GBEqJHO
uN4oBsLTn0Hv2jOA6xn3w8FfIkUGc4oueQp683O2ZJct3ownfVShMotBj35Dkp/+
XuAWIUo1K1mWDXqY7FRhQ/s+sxvM9oLfcpfZhFsCgYAYxJv72Rr9apmzQyCo90Ru
fbMdrnI+qo0QrV3m7uDMM7Y+jAvIW1bDVpkJGMVyGpF8ur9EFMXnDoDisdEmyamO
Lp/zsqcOxy7+EIbHhO4AVU85xbCygGU+OHN7nKPnLAV79hL5KNtvbQq9ic0zOip7
YIt5vAJsmvphr9pATz1VXQKBgQCnAYbs0TdgFhWoPNVI3AA+4/Bi/P91Gk4d5V3x
6SmkB4tj9lfvZMRFOhFrUqK+ZezgCmLBDSKcQyirT6A1GJFGbXbdaxPrG3JV9RZ8
Cancel
Decrypt password


Remote Desktop Connection
Remote Desktop
Connection
Computer:
13.201.72.107
User name:
MicrosoftAccount\Administrator
You will be asked for credentials when you connect.
Show Options

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2.Configure IIS Webserver on Windows Server in AWS

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Objective

An EC2 Windows instance is a virtual machine running the Microsoft Windows operating system on Amazon Web Services (AWS) EC2 (Elastic Compute Cloud). This allows you to run Windows-based applications, manage Windows Server configurations, and use various Microsoft software (like SQL Server, IIS, etc.) in the cloud without having to manage physical hardware.

IIS (Internet Information Services) is a web server software created by Microsoft for hosting websites and web applications on Windows Server operating systems. It provides a platform for building, deploying, and managing web-based applications. IIS supports multiple protocols, including HTTP, HTTPS, FTP, and more, and is tightly integrated with the Windows OS.

Common Use Cases of IIS:

1. Hosting Websites: IIS is often used by businesses and developers to host websites and web applications, particularly those built with ASP.NET.
2. Web Applications: For applications that require robust back-end support, IIS provides features like application pooling and security for web services (e.g., SOAP or RESTful APIs).
3. Secure File Sharing: IIS can be configured as an FTP server to securely share files between different locations.
4. Application Hosting in Enterprises: Many enterprise applications built on Microsoft technologies rely on IIS for deployment, such as SharePoint, Exchange Server, or Dynamics CRM.
5. API Hosting: IIS is commonly used to host APIs (e.g., RESTful services) built with .NET or other technologies.

Used Services in AWS

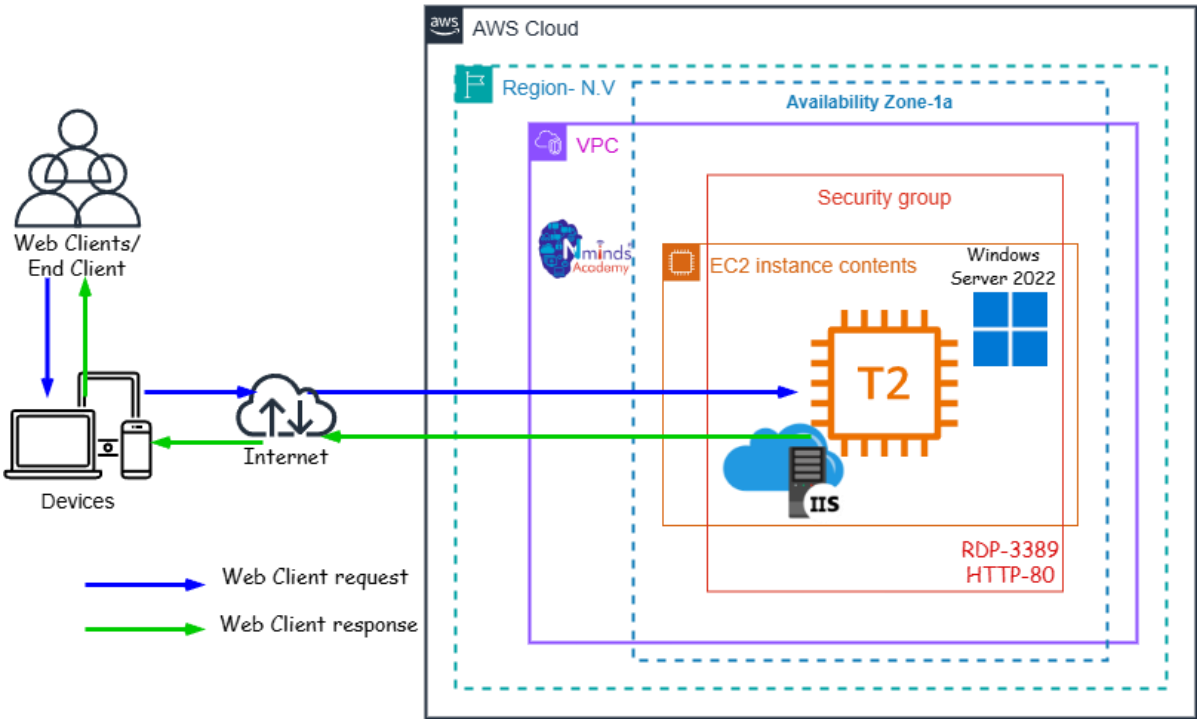
Used Services in Windows Server

Amazon EC2

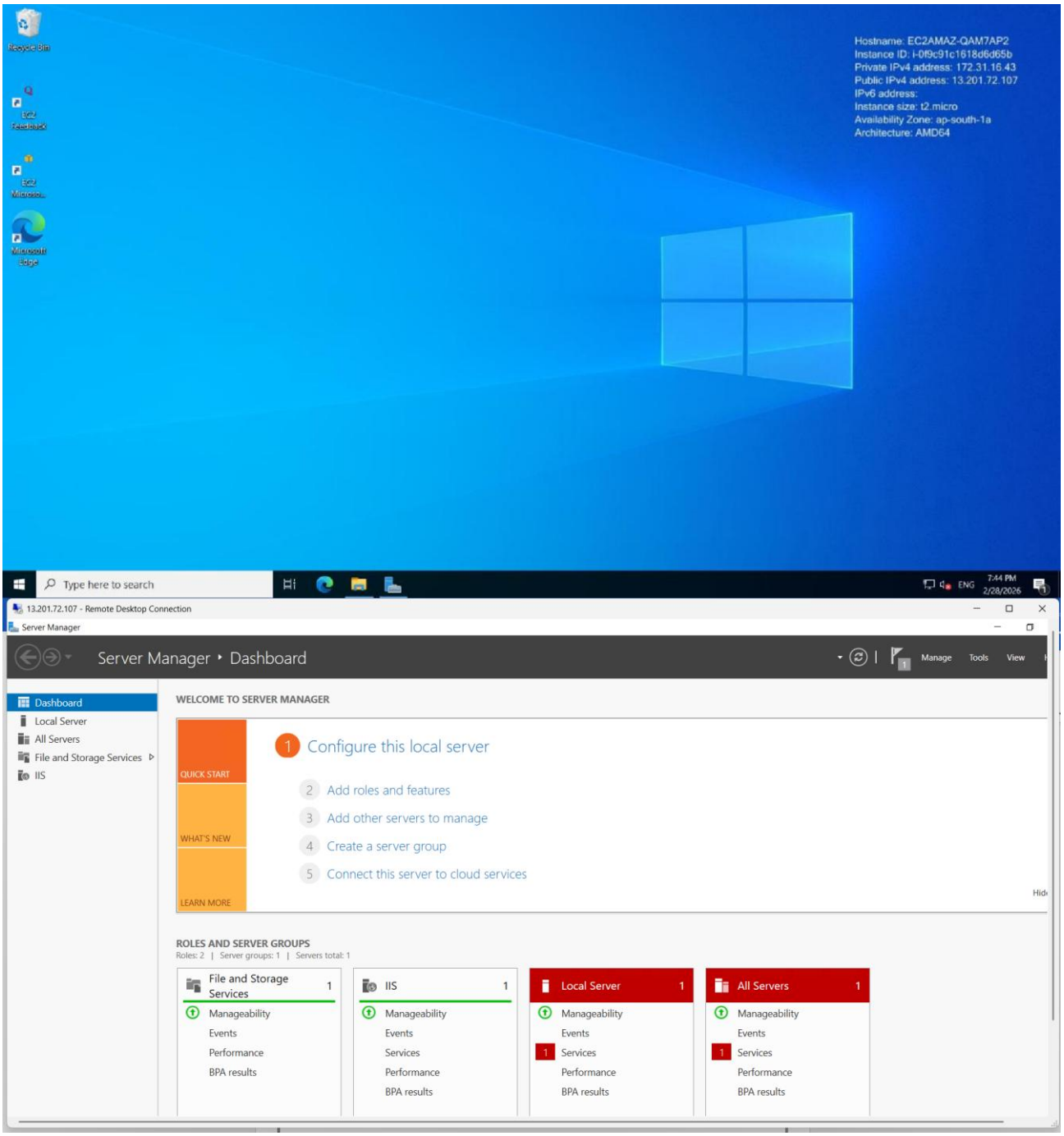
IIS- Internet Information Services

Topology

How to Configure the IIS web server in Windows Server with AWS EC2



Execution Tasks:



Select installation type

DESTINATION SERVER
EC2AMAZ-QAM7AP2

- Before You Begin
- Installation Type
- Server Selection
- Server Roles
- Features
- Confirmation
- Results

Select the installation type. You can install roles and features on a running physical computer or virtual machine, or on an offline virtual hard disk (VHD).

(●) **Role-based or feature-based installation**
Configure a single server by adding roles, role services, and features.

( ) **Remote Desktop Services installation**
Install required role services for Virtual Desktop Infrastructure (VDI) to create a virtual machine-based or session-based desktop deployment.

< Previous | Next > | Install | Cancel

Select server roles

DESTINATION SERVER
EC2AMAZ-QAM7AP2

- Before You Begin
- Installation Type
- Server Selection
- Server Roles
- Features
- Confirmation
- Results

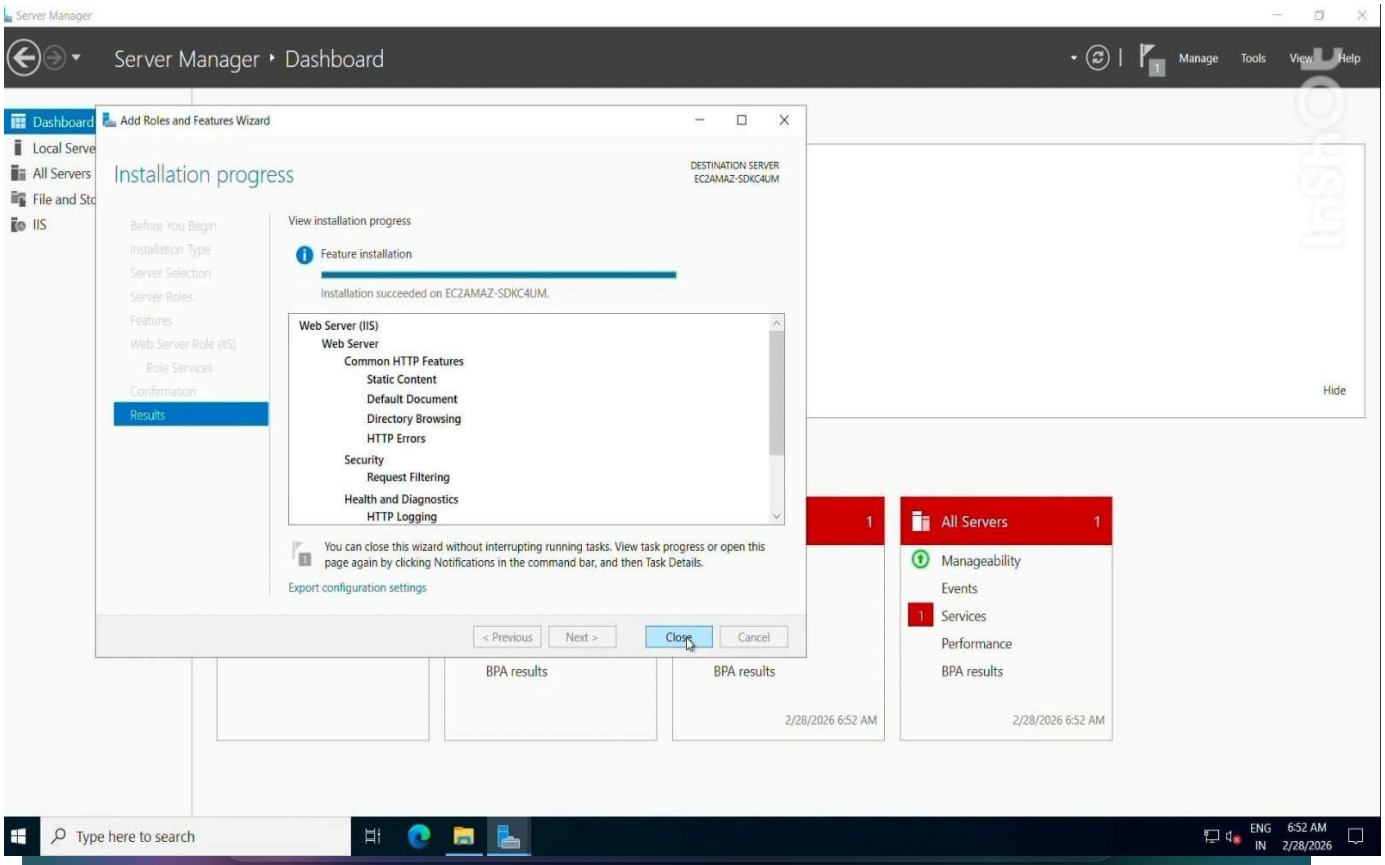
Select one or more roles to install on the selected server.

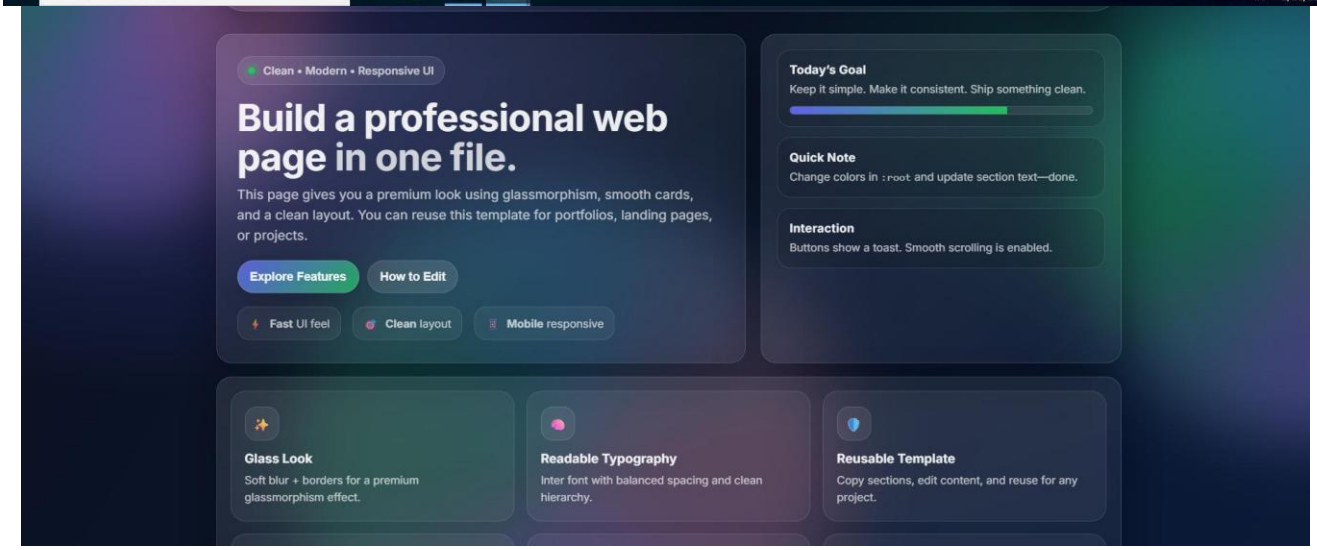
Roles

- [ ] Active Directory Domain Services
- [ ] Active Directory Federation Services
- [ ] Active Directory Lightweight Directory Services
- [ ] Active Directory Rights Management Services
- [ ] Device Health Attestation
- [ ] DHCP Server
- [ ] DNS Server
- [ ] Fax Server
- [■] File and Storage Services (1 of 12 installed)
- [ ] Host Guardian Service
- [ ] Hyper-V
- [ ] Network Controller
- [ ] Network Policy and Access Services
- [ ] Print and Document Services
- [ ] Remote Access
- [ ] Remote Desktop Services
- [ ] Volume Activation Services
- [■] Web Server (IIS) (8 of 43 installed)
- [ ] Windows Deployment Services
- [ ] Windows Server Update Services

Description

Active Directory Certificate Services (AD CS) is used to create certification authorities and related role services that allow you to issue and manage certificates used in a variety of applications.


Server Manager
Server Manager ‣ Dashboard
Manage
Tools
View
Help
Dashboard
Local Serve
All Servers
File and Sto
IIS
Add Roles and Features Wizard
Installation progress
DESTINATION SERVER
EC2AMAZ-SDKC4UM
Before You Begin
Installation Type
Server Selection
Server Roles
Features
Web Server Role (IIS)
Role Services
Confirmation
Results
View installation progress
Feature installation
Installation succeeded on EC2AMAZ-SDKC4UM.
Web Server (IIS)
Web Server
Common HTTP Features
Static Content
Default Document
Directory Browsing
HTTP Errors
Security
Request Filtering
Health and Diagnostics
HTTP Logging
You can close this wizard without interrupting running tasks. View task progress or open this page again by clicking Notifications in the command bar, and then Task Details.
Export configuration settings
< Previous
Next >
Close
Cancel
Hide
All Servers
Manageability
Events
Services
Performance
BPA results
BPA results
BPA results
2/28/2026 6:52 AM
2/28/2026 6:52 AM
Type here to search
ENG
IN
6:52 AM
2/28/2026


Clean • Modern • Responsive UI
Build a professional web page in one file.
This page gives you a premium look using glassmorphism, smooth cards, and a clean layout. You can reuse this template for portfolios, landing pages, or projects.
Explore Features
How to Edit
Fast UI feel
Clean layout
Mobile responsive
Today's Goal
Keep it simple. Make it consistent. Ship something clean.
Quick Note
Change colors in :root and update section text—done.
Interaction
Buttons show a toast. Smooth scrolling is enabled.
Glass Look
Soft blur + borders for a premium glassmorphism effect.
Readable Typography
Inter font with balanced spacing and clean hierarchy.
Reusable Template
Copy sections, edit content, and reuse for any project.

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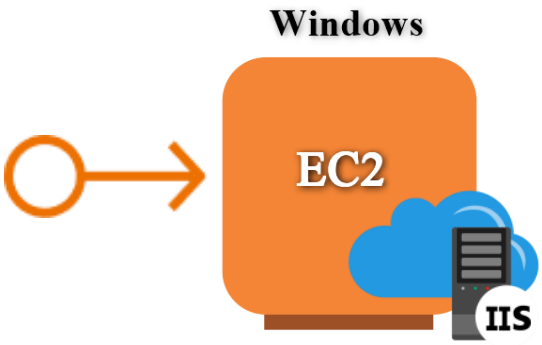
3.Configure Elastic IP Address to Windows Web Server in AWS



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Objective

An Elastic IP (EIP) in AWS is a static, public IP address designed for dynamic cloud computing. It is associated with your AWS account and can be quickly associated or disassociated with any EC2 instance in your account. This feature is particularly useful when you need to maintain a consistent IP address for your resources, even when you stop and start EC2 instances.

Common Use Cases for Elastic IP:

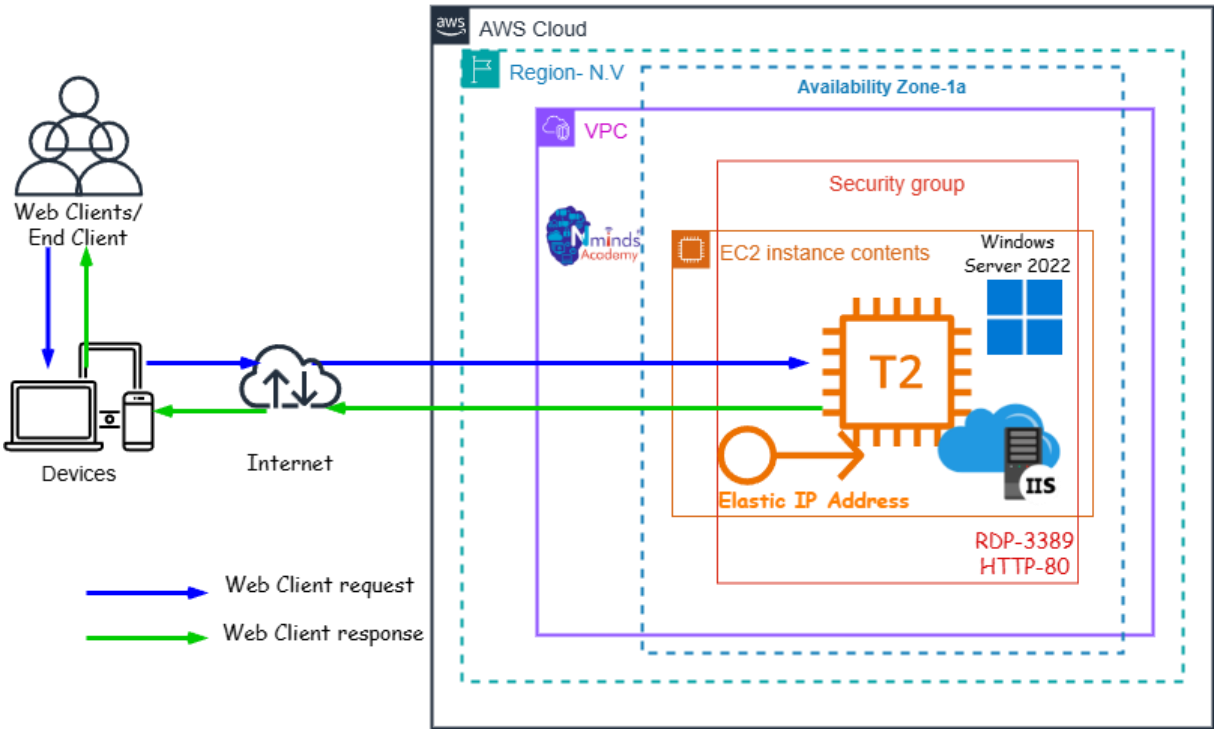
1. Highly Available Applications:
    - If you're running a service that requires high availability, you can use Elastic IPs to quickly reassign a static IP to a new instance if your primary instance fails, ensuring minimal downtime.
2. Web Servers:
    - If you host a website and need to ensure the IP address remains the same even if the underlying EC2 instance is restarted, an Elastic IP helps maintain this consistency.
3. Disaster Recovery:
    - Elastic IPs are useful for disaster recovery scenarios. If one instance goes down, you can quickly associate the EIP with a backup instance to ensure services are still accessible.

Best Practices:

- Use Elastic IPs only when necessary: Since they come with associated costs when unused, it's a good practice to release EIPs that are no longer required.
- Move IPs during instance failure: Instead of keeping an EIP permanently attached, use it as a failover method, reassociating it to a new instance when needed.
- Monitor EIP Usage: Periodically review your usage to ensure you're not paying for unnecessary Elastic IP addresses.

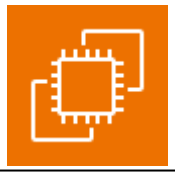
Topology

How to Configure the Elastic IP Address to Windows Web Server with AWS EC2



Used Services in AWS

Used Services in Windows Server

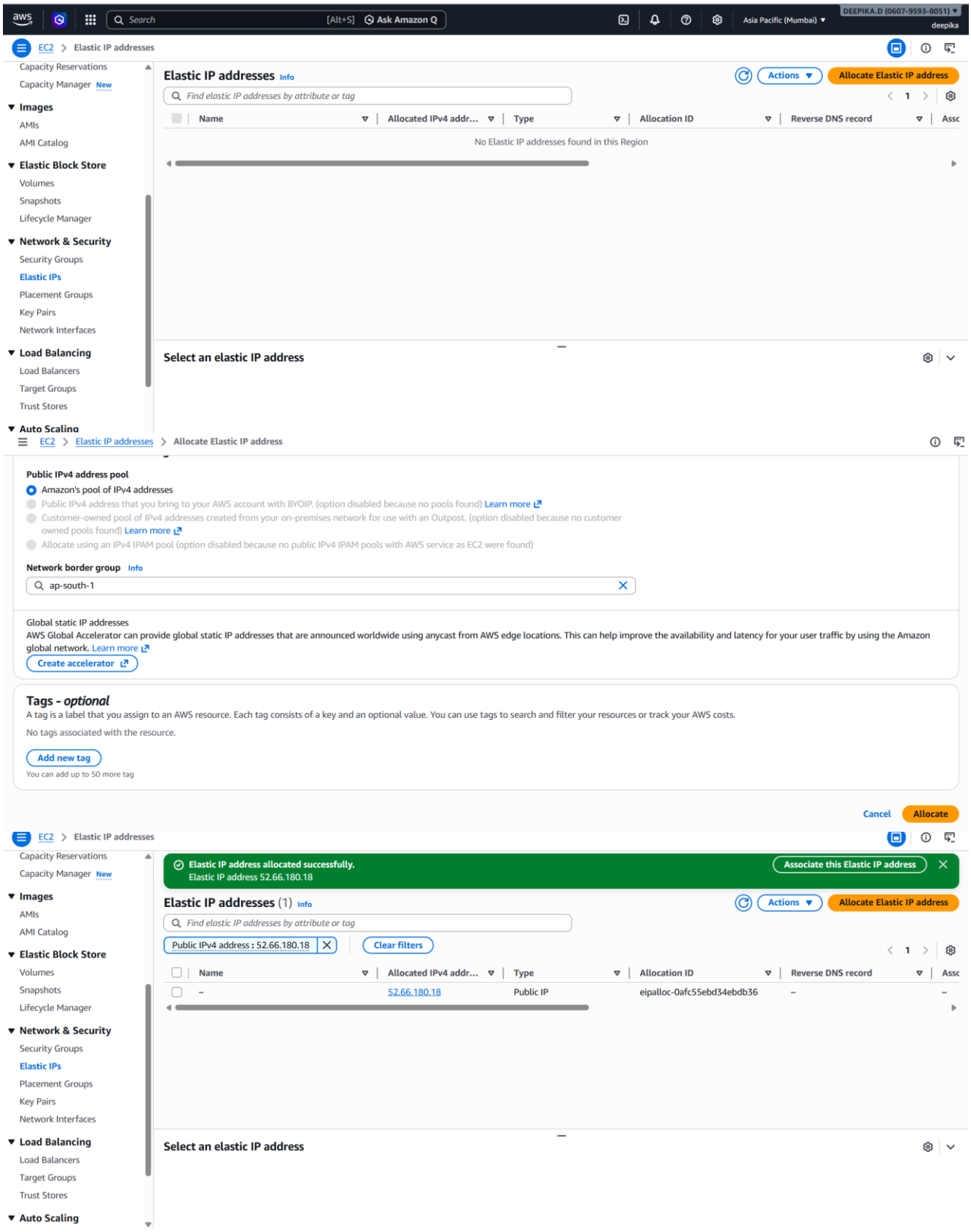


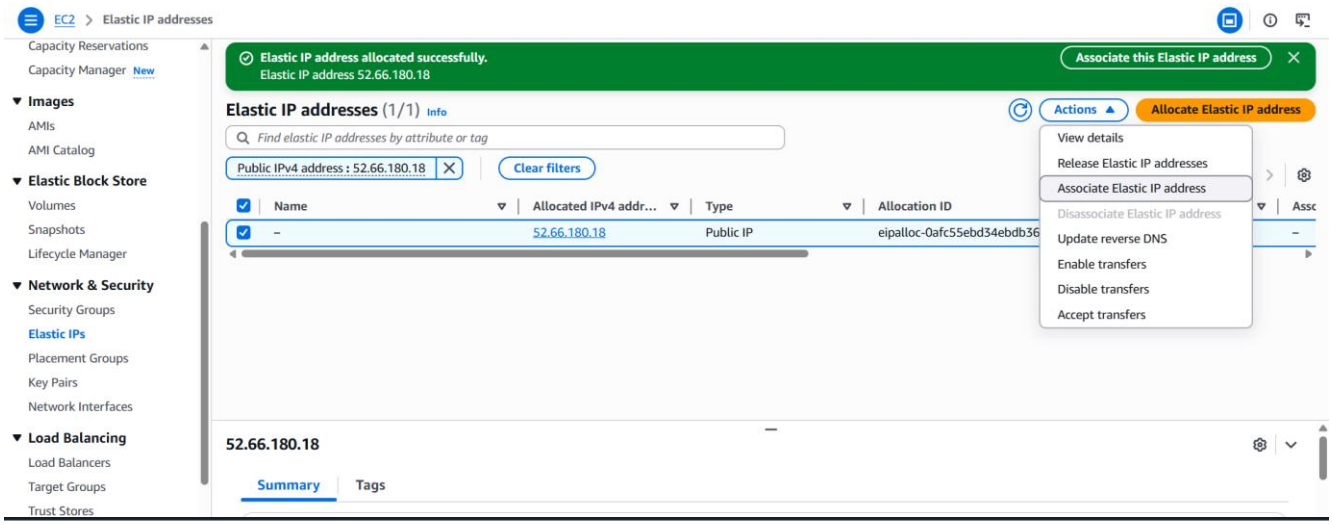
Amazon EC2

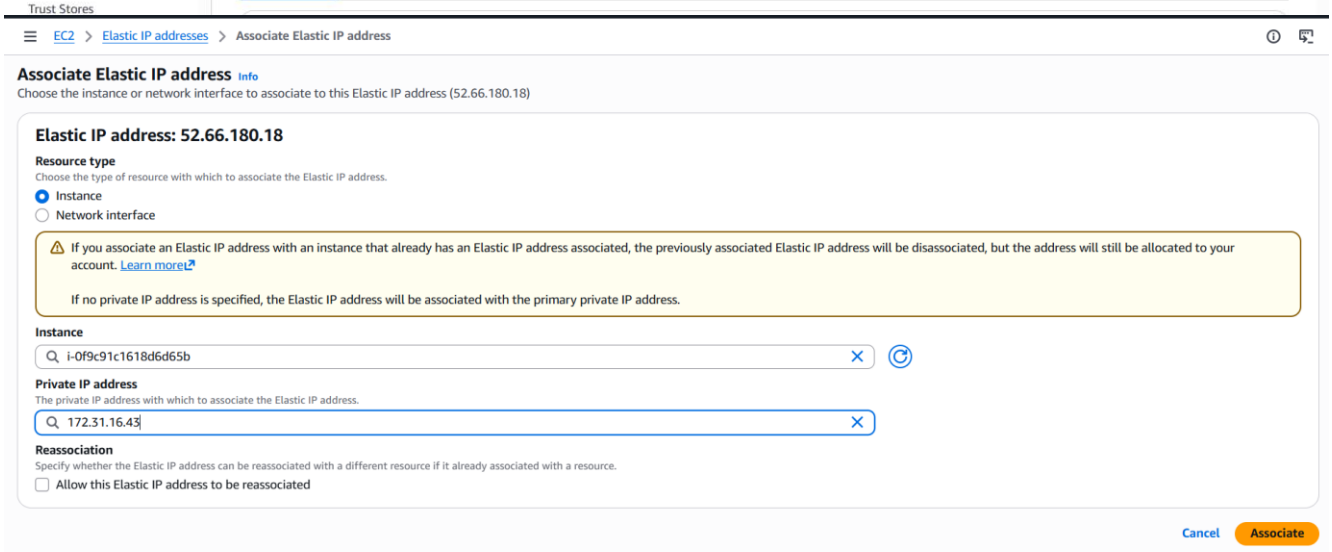
IIS- Internet Information Services

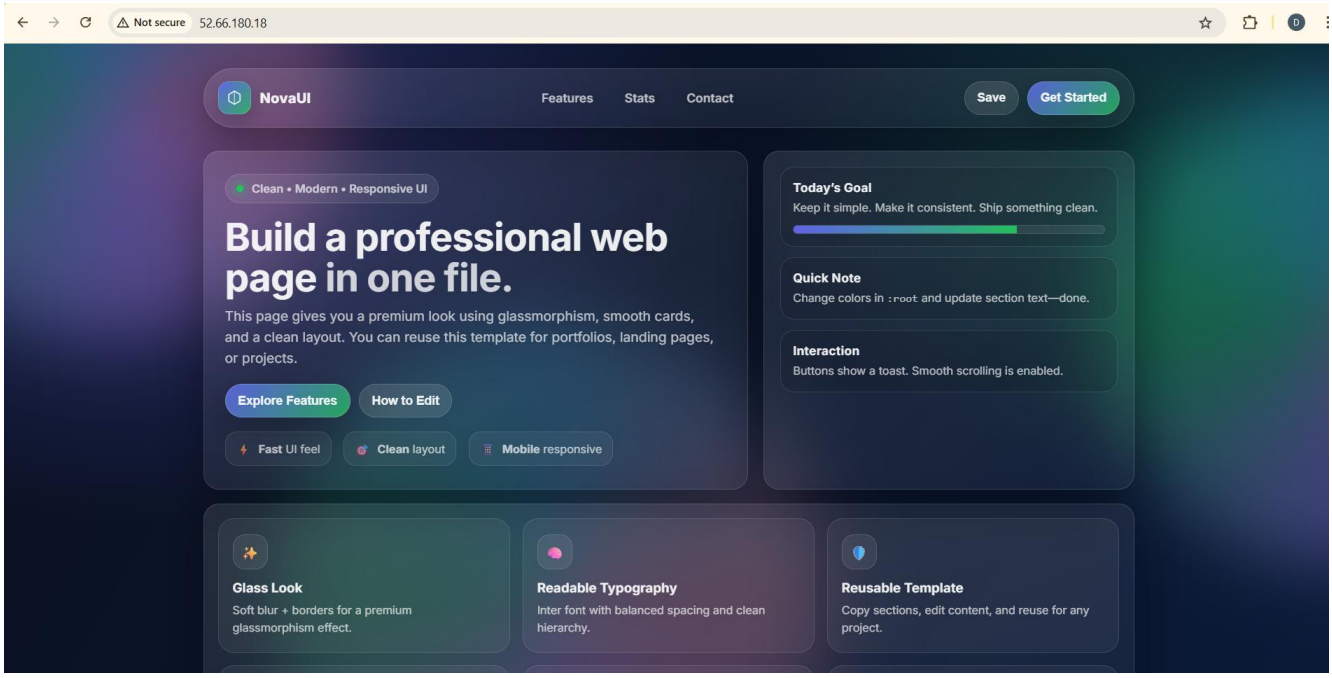
Amazon Elastic IP Address

Execution Tasks:




EC2 > Elastic IP addresses
Capacity Reservations
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces
Load Balancing
Load Balancers
Target Groups
Trust Stores
Elastic IP address allocated successfully.
Elastic IP address 52.66.180.18
Associate this Elastic IP address
Elastic IP addresses (1/1) Info
Actions
Allocate Elastic IP address
Find elastic IP addresses by attribute or tag
Public IPv4 address : 52.66.180.18
Clear filters
View details
Release Elastic IP addresses
Associate Elastic IP address
Disassociate Elastic IP address
Update reverse DNS
Enable transfers
Disable transfers
Accept transfers
Name
Allocated IPv4 addr...
Type
Allocation ID
52.66.180.18
Public IP
eipalloc-0afc55ebd34ebdb36
52.66.180.18
Summary
Tags


EC2 > Elastic IP addresses > Associate Elastic IP address
Associate Elastic IP address Info
Choose the instance or network interface to associate to this Elastic IP address (52.66.180.18)
Elastic IP address: 52.66.180.18
Resource type
Choose the type of resource with which to associate the Elastic IP address.
Instance
Network interface
If you associate an Elastic IP address with an instance that already has an Elastic IP address associated, the previously associated Elastic IP address will be disassociated, but the address will still be allocated to your account. Learn more
If no private IP address is specified, the Elastic IP address will be associated with the primary private IP address.
Instance
i-0f9c91c1618d6d65b
Private IP address
The private IP address with which to associate the Elastic IP address.
172.31.16.43
Reassociation
Specify whether the Elastic IP address can be reassociated with a different resource if it already associated with a resource.
Allow this Elastic IP address to be reassociated
Cancel
Associate


Not secure
52.66.180.18
NovaUI
Features
Stats
Contact
Save
Get Started
Clean • Modern • Responsive UI
Build a professional web page in one file.
This page gives you a premium look using glassmorphism, smooth cards, and a clean layout. You can reuse this template for portfolios, landing pages, or projects.
Explore Features
How to Edit
Fast UI feel
Clean layout
Mobile responsive
Today's Goal
Keep it simple. Make it consistent. Ship something clean.
Quick Note
Change colors in :root and update section text—done.
Interaction
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Inter font with balanced spacing and clean hierarchy.
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4. Create an EC2 Linux Instance in AWS

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Objective

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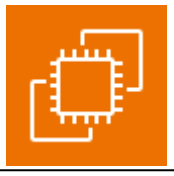
An EC2 Linux instance is a virtual machine running the Linux operating system on Amazon Web Services (AWS) EC2 (Elastic Compute Cloud). This allows you to run Windows-based applications, manage Windows Server configurations, and use various Microsoft software (like SQL Server, Apache,FileServer etc.) in the cloud without having to manage physical hardware.

Use Cases for EC2 Linux Instances:

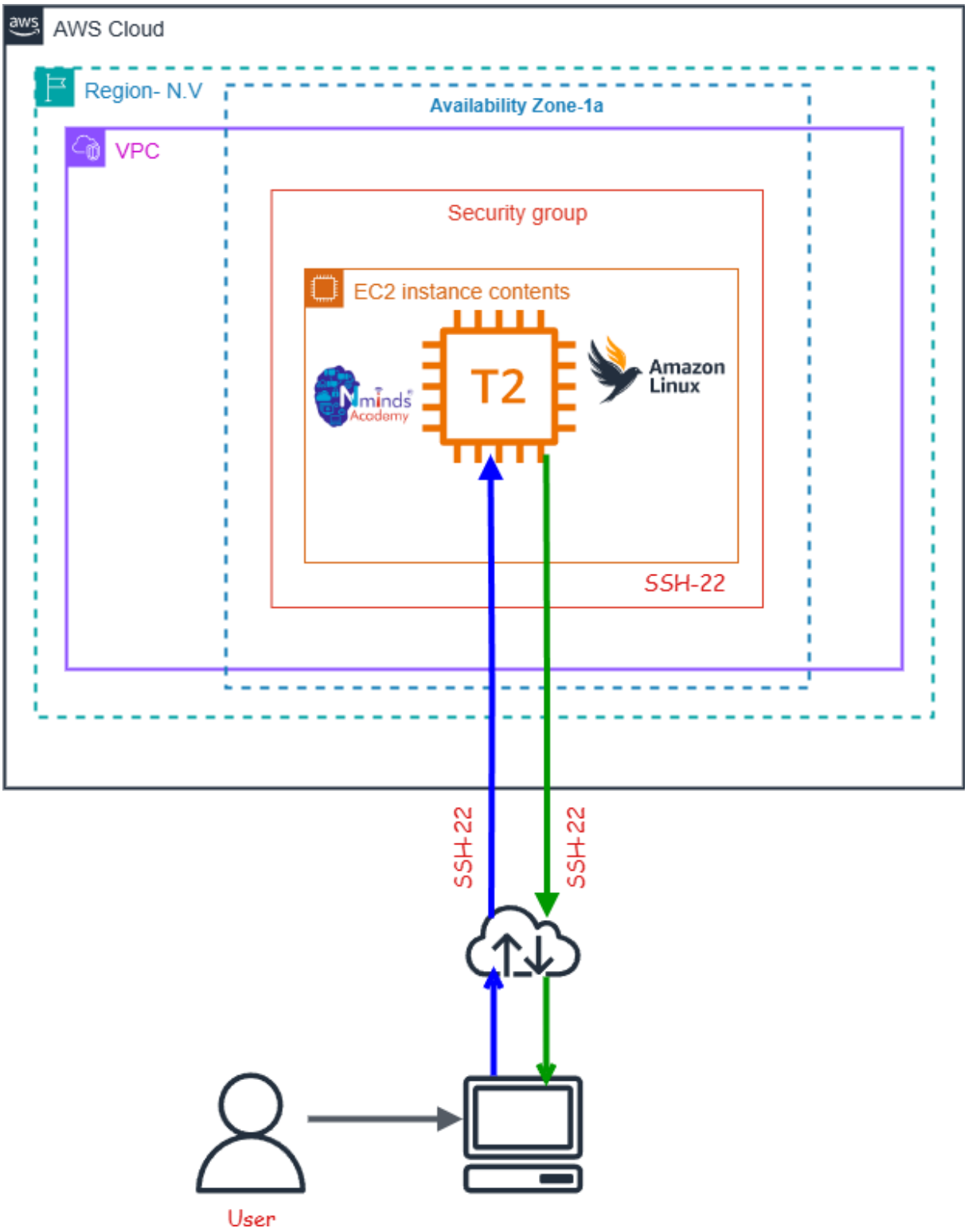
- Web and Application Hosting: Run Apache web servers or other Linux-based web applications in the cloud.
- Database Servers: Host SQL Server databases with EC2 Linux instances, using SSH or EC2 for more control.
- Remote Desktop Applications: Set up SSH (Remote Desktop Services) for remote access to Linux desktops in the cloud.
- Development and Testing: Create development environments with linux OS to test php,MySql or other Linux-based applications.

Used Services in AWS

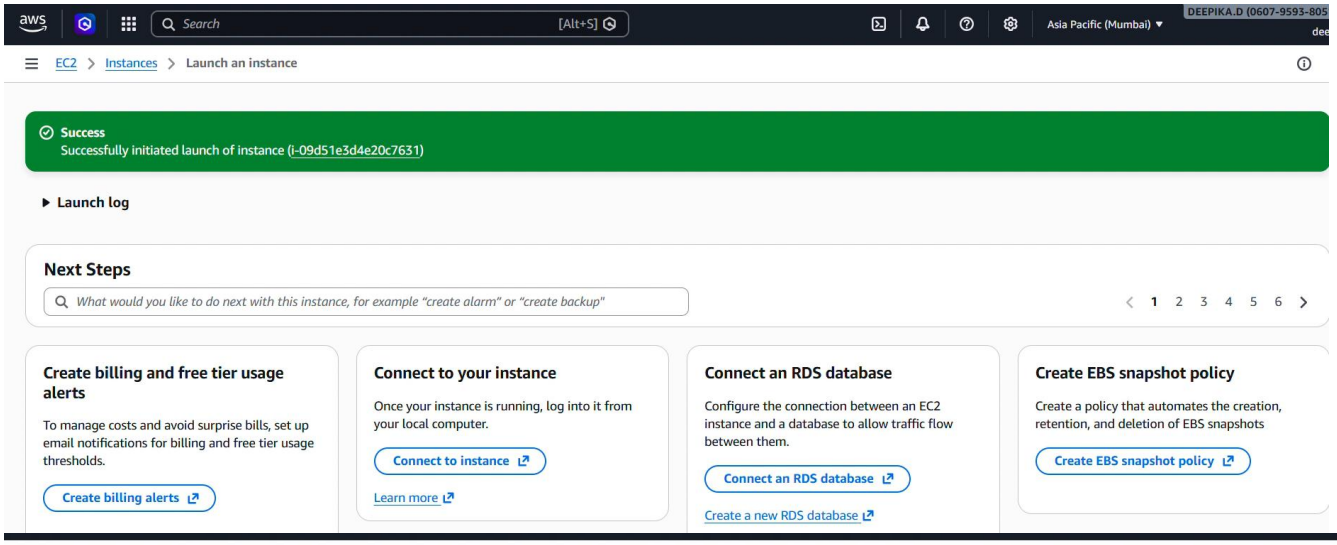


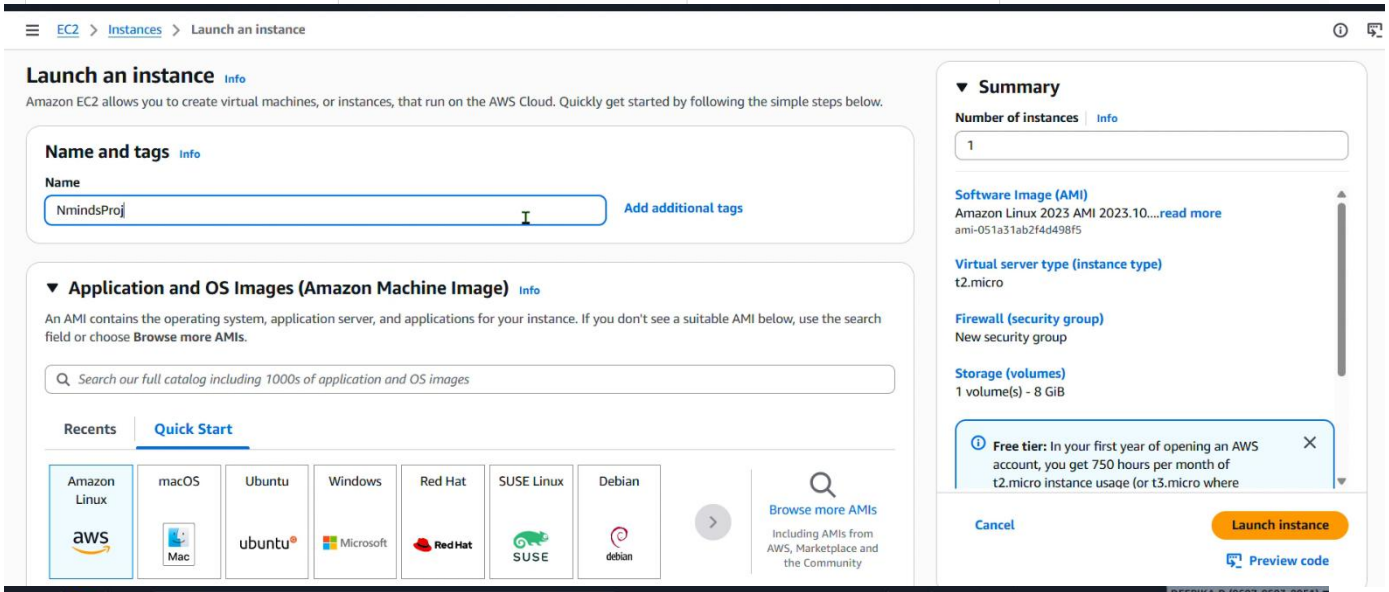
Amazon EC2

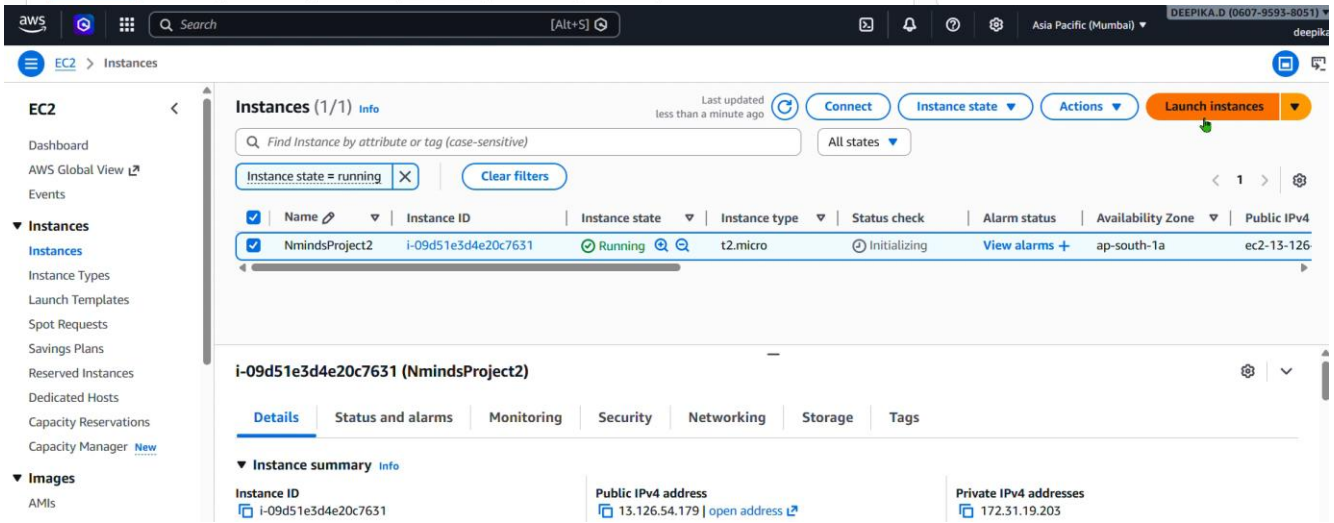
Topology



Execution Tasks:


Search
[Alt+S]
Asia Pacific (Mumbai)
DEEPIKA.D (0607-9593-805
EC2 > Instances > Launch an instance
Success
Successfully initiated launch of instance (i-09d51e3d4e20c7631)
Launch log
Next Steps
What would you like to do next with this instance, for example "create alarm" or "create backup"
1 2 3 4 5 6
Create billing and free tier usage alerts
To manage costs and avoid surprise bills, set up email notifications for billing and free tier usage thresholds.
Create billing alerts
Connect to your instance
Once your instance is running, log into it from your local computer.
Connect to instance
Learn more
Connect an RDS database
Configure the connection between an EC2 instance and a database to allow traffic flow between them.
Connect an RDS database
Create a new RDS database
Create EBS snapshot policy
Create a policy that automates the creation, retention, and deletion of EBS snapshots
Create EBS snapshot policy


EC2 > Instances > Launch an instance
Launch an instance Info
Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.
Name and tags Info
Name
NmindsProj
Add additional tags
Application and OS Images (Amazon Machine Image) Info
An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose Browse more AMIs.
Search our full catalog including 1000s of application and OS images
Recents
Quick Start
Amazon Linux
macOS
Ubuntu
Windows
Red Hat
SUSE Linux
Debian
Browse more AMIs
Including AMIs from AWS, Marketplace and the Community
Summary
Number of instances Info
1
Software Image (AMI)
Amazon Linux 2023 AMI 2023.10....read more
ami-051a31ab2f4d498f5
Virtual server type (instance type)
t2.micro
Firewall (security group)
New security group
Storage (volumes)
1 volume(s) - 8 GiB
Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where
Cancel
Launch instance
Preview code


Search
[Alt+S]
Asia Pacific (Mumbai)
DEEPIKA.D (0607-9593-8051)
EC2 > Instances
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager New
Images
AMIs
Instances (1/1) Info
Last updated less than a minute ago
Connect
Instance state
Actions
Launch instances
Find Instance by attribute or tag (case-sensitive)
All states
Instance state = running
Clear filters
Name
Instance ID
Instance state
Instance type
Status check
Alarm status
Availability Zone
Public IPv4
NmindsProject2
i-09d51e3d4e20c7631
Running
t2.micro
Initializing
View alarms +
ap-south-1a
ec2-13-126
i-09d51e3d4e20c7631 (NmindsProject2)
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags
Instance summary Info
Instance ID
i-09d51e3d4e20c7631
Public IPv4 address
13.126.54.179 | open address
Private IPv4 addresses
172.31.19.203

```
(c) Microsoft Corporation. All rights reserved.

C:\Users\deepi>cd downloads

C:\Users\deepi\Downloads>ssh -i NmindsProject1key.pem ec2-user@13.127.122.14
The authenticity of host '13.127.122.14 (13.127.122.14)' can't be established.
ED25519 key fingerprint is SHA256:VbMU5CjJ8KBpVfl0+hK3ChX4EadVyBCXHvjaGm8rrqU.
This host key is known by the following other names/addresses:
    C:\Users\deepi/.ssh/known_hosts:16: 13.126.54.179
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '13.127.122.14' (ED25519) to the list of known hosts.
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Sat Feb 28 16:50:20 2026 from 121.200.55.210
[ec2-user@ip-172-31-19-203 ~]$ sudo -i
[root@ip-172-31-19-203 ~]# systemctl status httpd
● httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
     Active: active (running) since Sat 2026-02-28 16:42:29 UTC; 20min ago
       Docs: man:httpd.service(8)
   Main PID: 25972 (httpd)
     Status: "Total requests: 7; Idle/Busy workers 100/0;Requests/sec: 0.00583; Bytes served/sec:  53 B/sec"
      Tasks: 230 (limit: 1120)
     Memory: 16.5M
        CPU: 850ms
     CGroup: /system.slice/httpd.service
             ├─25972 /usr/sbin/httpd -DFOREGROUND
             ├─25989 /usr/sbin/httpd -DFOREGROUND
             ├─25991 /usr/sbin/httpd -DFOREGROUND
             ├─25992 /usr/sbin/httpd -DFOREGROUND
             ├─25993 /usr/sbin/httpd -DFOREGROUND
             └─26277 /usr/sbin/httpd -DFOREGROUND
```

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5. Configure Apache Web Server on Linux Instance in AWS

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Objective

EC2 (Elastic Compute Cloud) is a core service provided by Amazon Web Services (AWS) that enables users to rent virtual servers in the cloud to run applications, store data, or handle computational workloads.

An EC2 Linux instance is a virtual machine running the Linux operating system on Amazon Web Services (AWS) EC2 (Elastic Compute Cloud). This allows you to run Windows-based applications, manage Windows Server configurations, and use various Microsoft software (like SQL Server, Apache,FileServer etc.) in the cloud without having to manage physical hardware.

Use Cases for EC2 Linux Instances:

- Web and Application Hosting: Run Apache web servers or other Linux-based web applications in the cloud.
- Database Servers: Host SQL Server databases with EC2 Linux instances, using SSH or EC2 for more control.
- Remote Desktop Applications: Set up SSH (Remote Desktop Services) for remote access to Linux desktops in the cloud.
- Development and Testing: Create development environments with linux OS to test php,MySql or other Linux-based applications.

Used Services in AWS

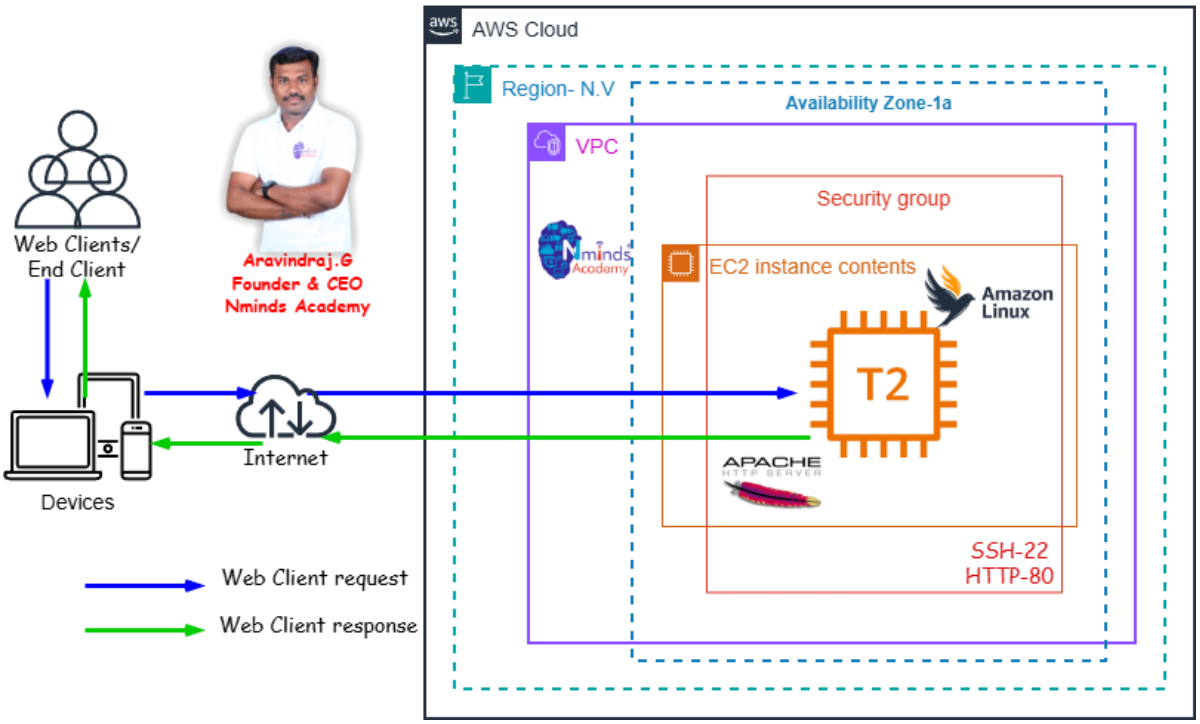
Amazon EC2

Used Services in Linux Server

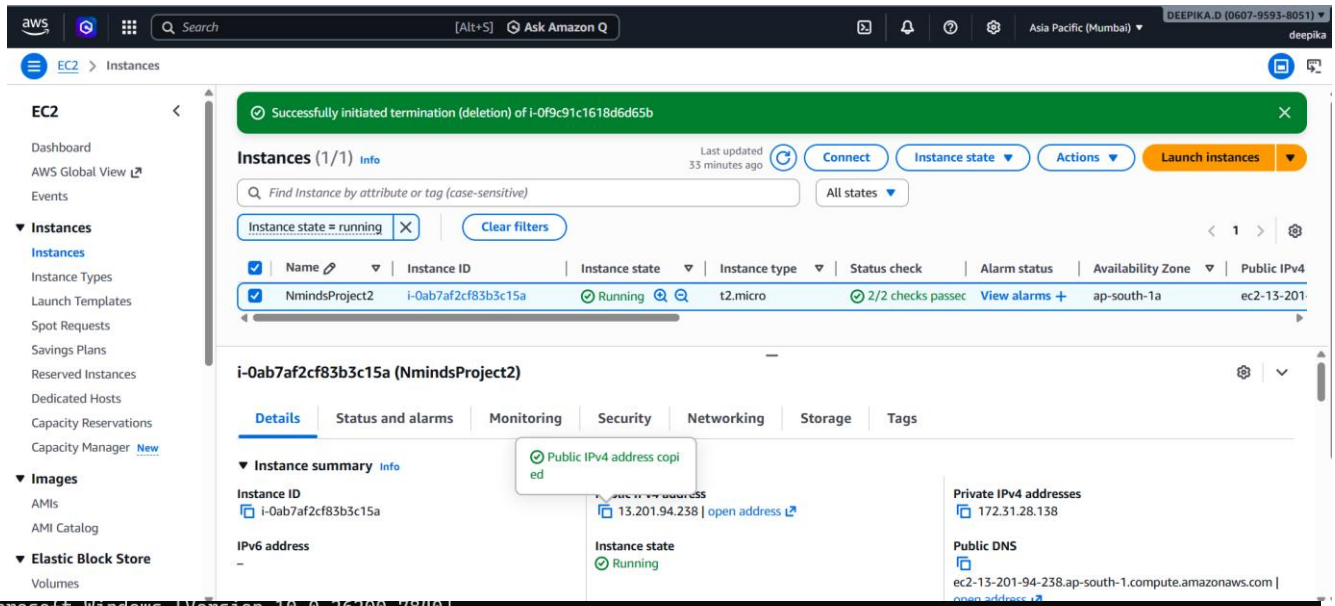
Apache Webserver

Topology

How to Configure the Apache web server in Linux Server with AWS EC2



Execution Tasks:




```
Microsoft Windows [Version 10.0.26200.7840]
(c) Microsoft Corporation. All rights reserved.

C:\Users\deepi>cd downloads

C:\Users\deepi\Downloads>ssh -i NmindsProject1key.pem ec2-user@13.201.94.238
       #_
       ####_        Amazon Linux 2023
      \_#####\
          \###|
            \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
             V~' '->
              /
            _/ _/
          _/m/'
Last login: Sun Mar  1 15:15:20 2026 from 103.183.240.250
[ec2-user@ip-172-31-28-138 ~]$ sudo -i
[root@ip-172-31-28-138 ~]# yum update -y
yum install hAmazon Linux 2023 Kernel Livepatch repos     [===                                  ] ---  B/s |   0  B
Amazon Linux 2023 Kernel Livepatch repository                                    221 kB/s |  31 kB     00:00
Dependencies resolved.
Nothing to do.
Complete!
[root@ip-172-31-28-138 ~]# yum install httpd -y
Last metadata expiration check: 0:00:20 ago on Sun Mar  1 15:43:13 2026.
Dependencies resolved.
================================================================================================================
 Package                        Architecture       Version                              Repository              Size
================================================================================================================
```

```
Complete!
[root@ip-172-31-28-138 ~]# systemctl start httpd
[root@ip-172-31-28-138 ~]# systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[root@ip-172-31-28-138 ~]# systemctl status httpd
● httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
     Active: active (running) since Sun 2026-03-01 15:43:50 UTC; 17s ago
       Docs: man:httpd.service(8)
   Main PID: 27067 (httpd)
     Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes served/sec:   0 B/sec"
      Tasks: 177 (limit: 1120)
     Memory: 12.9M
        CPU: 63ms
     CGroup: /system.slice/httpd.service
             ├─27067 /usr/sbin/httpd -DFOREGROUND
             ├─27069 /usr/sbin/httpd -DFOREGROUND
             ├─27070 /usr/sbin/httpd -DFOREGROUND
             ├─27071 /usr/sbin/httpd -DFOREGROUND
             └─27072 /usr/sbin/httpd -DFOREGROUND

Mar 01 15:43:50 ip-172-31-28-138.ap-south-1.compute.internal systemd[1]: Starting httpd.service - The Apache HTTP Serve>
Mar 01 15:43:50 ip-172-31-28-138.ap-south-1.compute.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Mar 01 15:43:50 ip-172-31-28-138.ap-south-1.compute.internal httpd[27067]: Server configured, listening on: port 80
lines 1-19/19 (END)
```

Feed | LinkedIn | Instances | EC2 | ap-south-1 | It works! Apache httpd

Not secure 13.201.94.238

It works!

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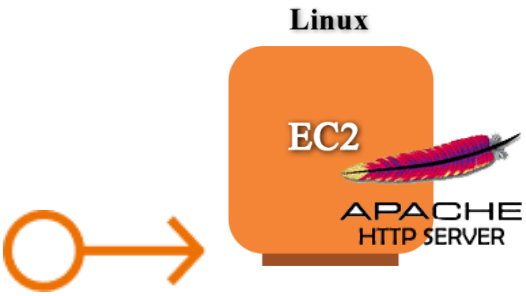
6. Configure Elastic IP Address to Windows Web Server in AWS



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Objective

An Elastic IP (EIP) in AWS is a static, public IP address designed for dynamic cloud computing. It is associated with your AWS account and can be quickly associated or disassociated with any EC2 instance in your account. This feature is particularly useful when you need to maintain a consistent IP address for your resources, even when you stop and start EC2 instances.

Common Use Cases for Elastic IP:

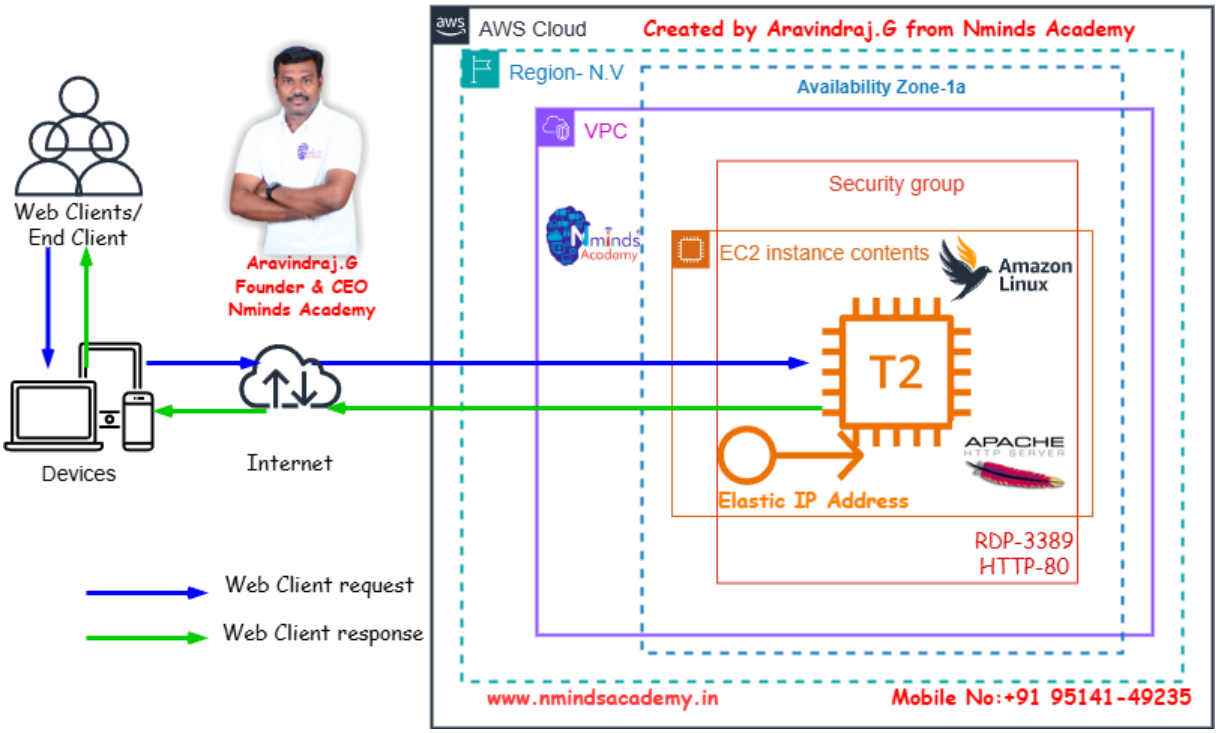
1. Highly Available Applications:
   - If you're running a service that requires high availability, you can use Elastic IPs to quickly reassign a static IP to a new instance if your primary instance fails, ensuring minimal downtime.
2. Web Servers:
   - If you host a website and need to ensure the IP address remains the same even if the underlying EC2 instance is restarted, an Elastic IP helps maintain this consistency.
3. Disaster Recovery:
   - Elastic IPs are useful for disaster recovery scenarios. If one instance goes down, you can quickly associate the EIP with a backup instance to ensure services are still accessible.

Best Practices:

- Use Elastic IPs only when necessary: Since they come with associated costs when unused, it's a good practice to release EIPs that are no longer required.
- Move IPs during instance failure: Instead of keeping an EIP permanently attached, use it as a failover method, reassociating it to a new instance when needed.
- Monitor EIP Usage: Periodically review your usage to ensure you're not paying for unnecessary Elastic IP addresses.

Topology

How to Configure the Elastic IP Address to Linux Web Server with AWS EC2



Used Services in AWS

Used Services in Windows Server

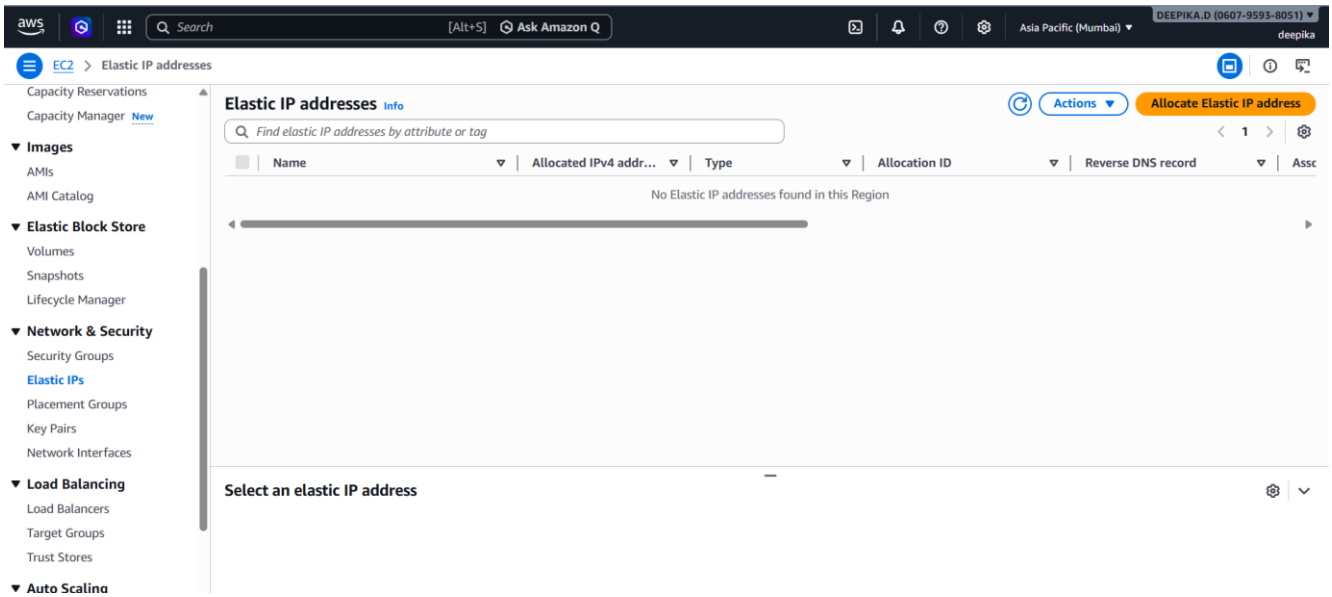
Amazon EC2

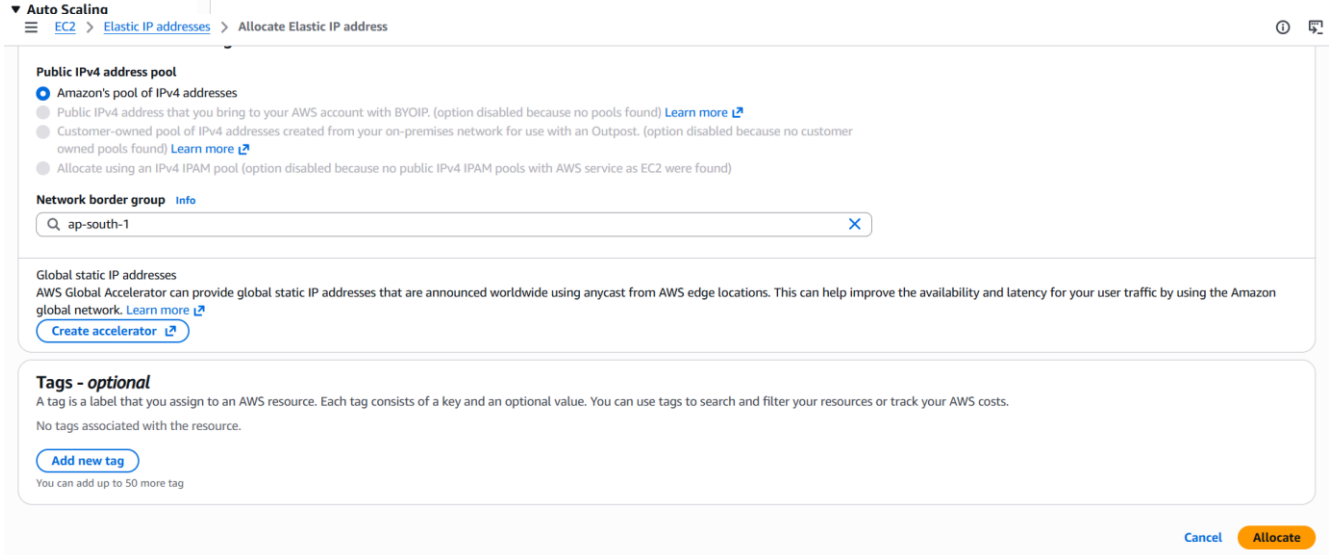
APACHE HTTP SERVER

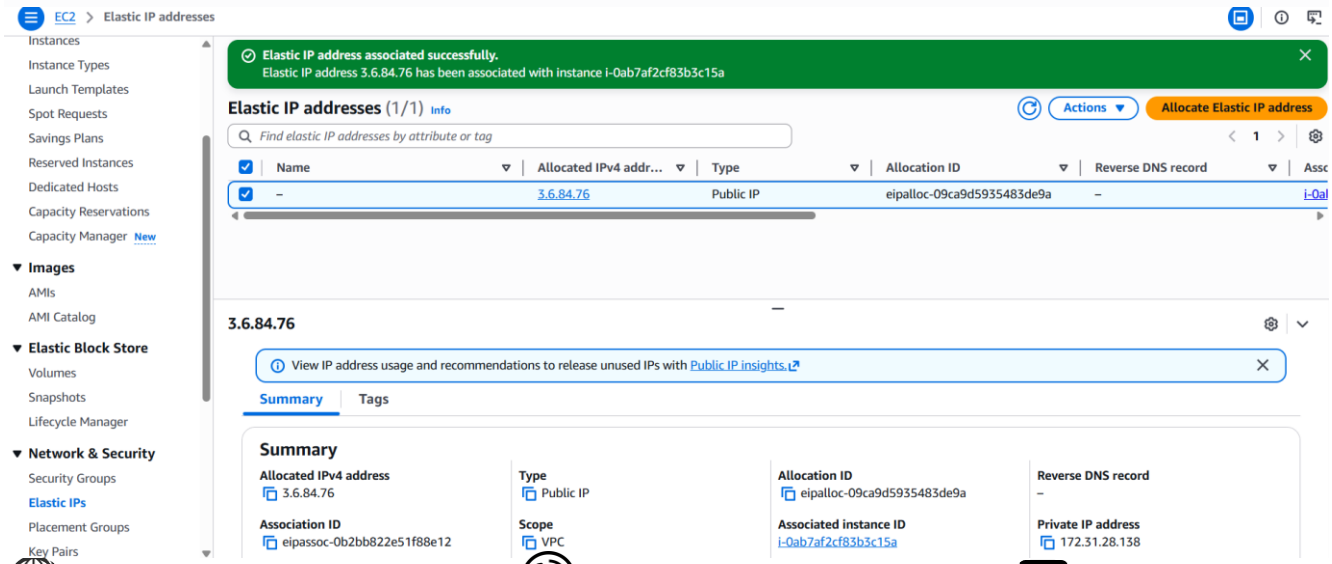
Apache Webserver

Amazon Elastic IP Address

Execution Tasks:







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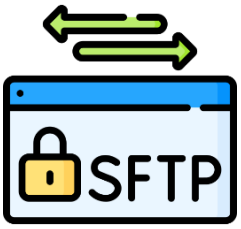
7.Configure SFTP to Linux Web Server in AWS



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Objective

EC2 (Elastic Compute Cloud) is a core service provided by Amazon Web Services (AWS) that enables users to rent virtual servers in the cloud to run applications, store data, or handle computational workloads.

Use Cases for EC2 Linux Instances:

- Web and Application Hosting: Run Apache web servers or other Linux- based web applications in the cloud.
- Database Servers: Host SQL Server databases with EC2 Linux instances, using SSH or EC2 for more control.
- Remote Desktop Applications: Set up SSH (Remote Desktop Services) for remote access to Linux desktops in the cloud.
- Development and Testing: Create development environments with linux OS to test php,MySql or other Linux-based applications.

SFTP- Secure File Transfer Protocol

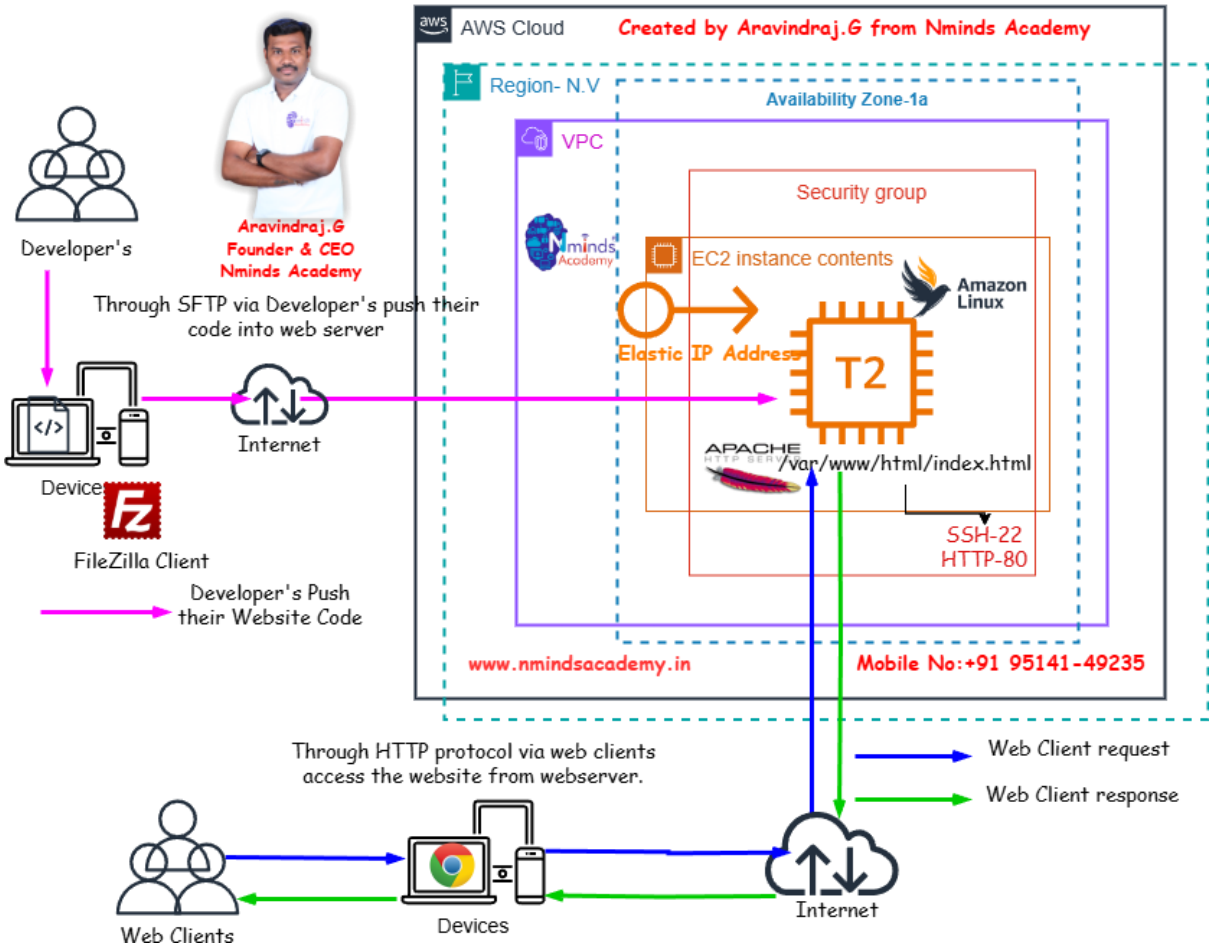
SFTP (Secure File Transfer Protocol) is a secure network protocol that allows users to transfer files over a secure, encrypted connection. It is a more secure alternative to FTP (File Transfer Protocol) and is commonly used to upload, download, and manage files on remote servers.

Here are the key uses and advantages of SFTP:

1. File Transfer: SFTP allows secure transfer of files between a local and a remote computer over a network. It ensures that the data is encrypted during transmission, protecting it from eavesdropping or tampering.
2. File Management: Besides transferring files, SFTP also allows users to perform file management tasks such as renaming, deleting, and moving files on the remote server.
3. Secure Authentication: SFTP uses SSH (Secure Shell) for authentication, which provides a stronger level of security compared to FTP's traditional username/password system. It often supports public key authentication, adding an extra layer of security.
4. Encryption: The main advantage of SFTP over FTP is encryption. All data sent over SFTP is encrypted, ensuring that sensitive information (like passwords and files) is protected during transmission.
5. Firewall-Friendly: SFTP operates over a single port (usually port 22), making it easier to use behind firewalls and NAT (Network Address Translation) devices compared to FTP, which uses multiple ports.

Topology

How to Configure Developer's push their website code by using SFTP Protocol in Linux Server with AWS EC2 with Elastic IP Address



Used Services in AWS

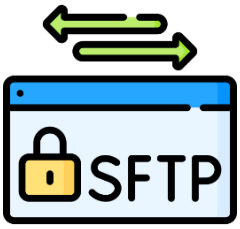
Used Services in Windows Server

Amazon EC2

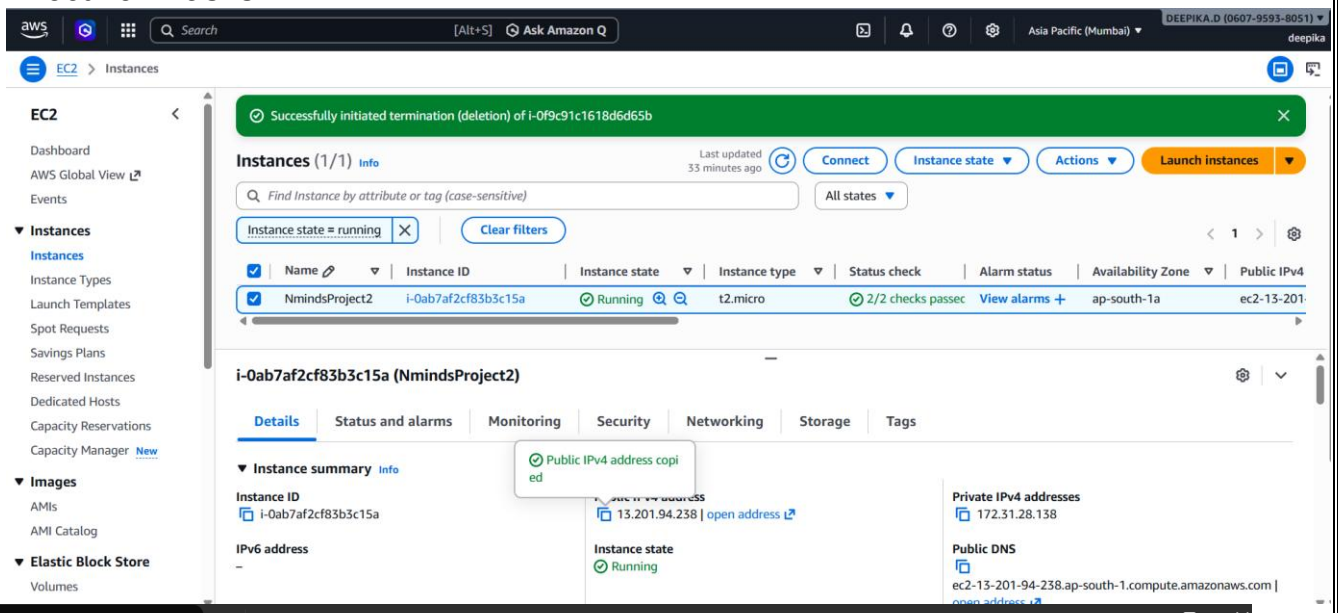
APACHE HTTP SERVER

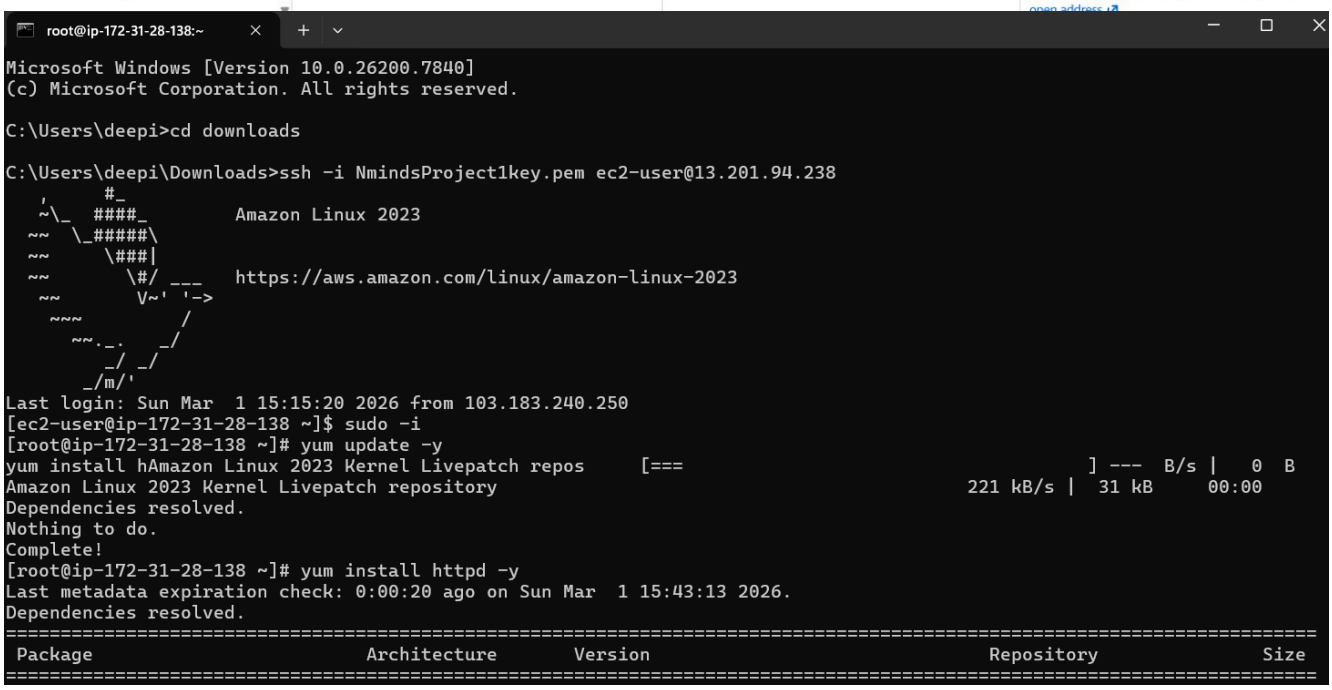
Apache Webserver

Amazon Elastic IP Address



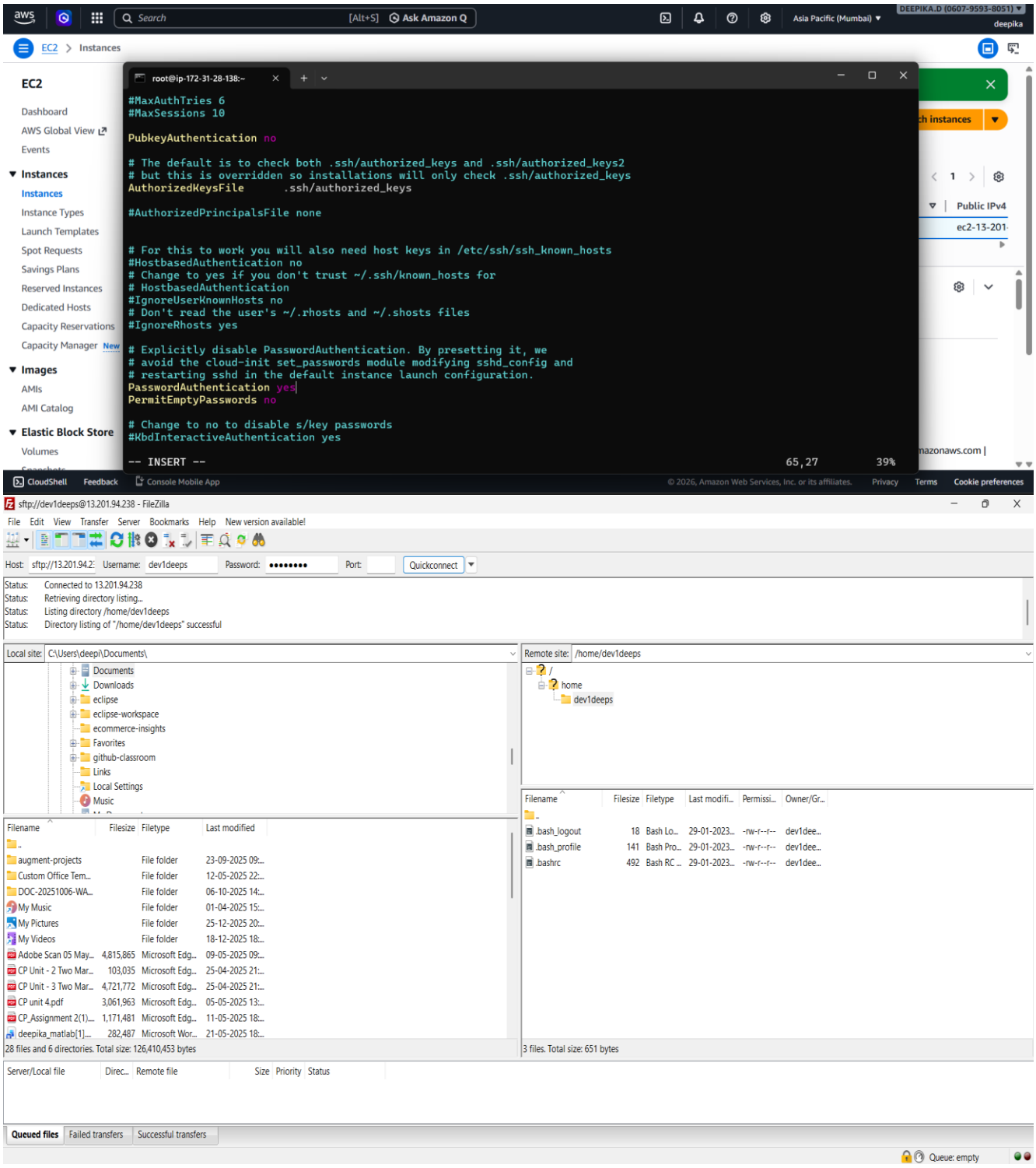
Execution Tasks:

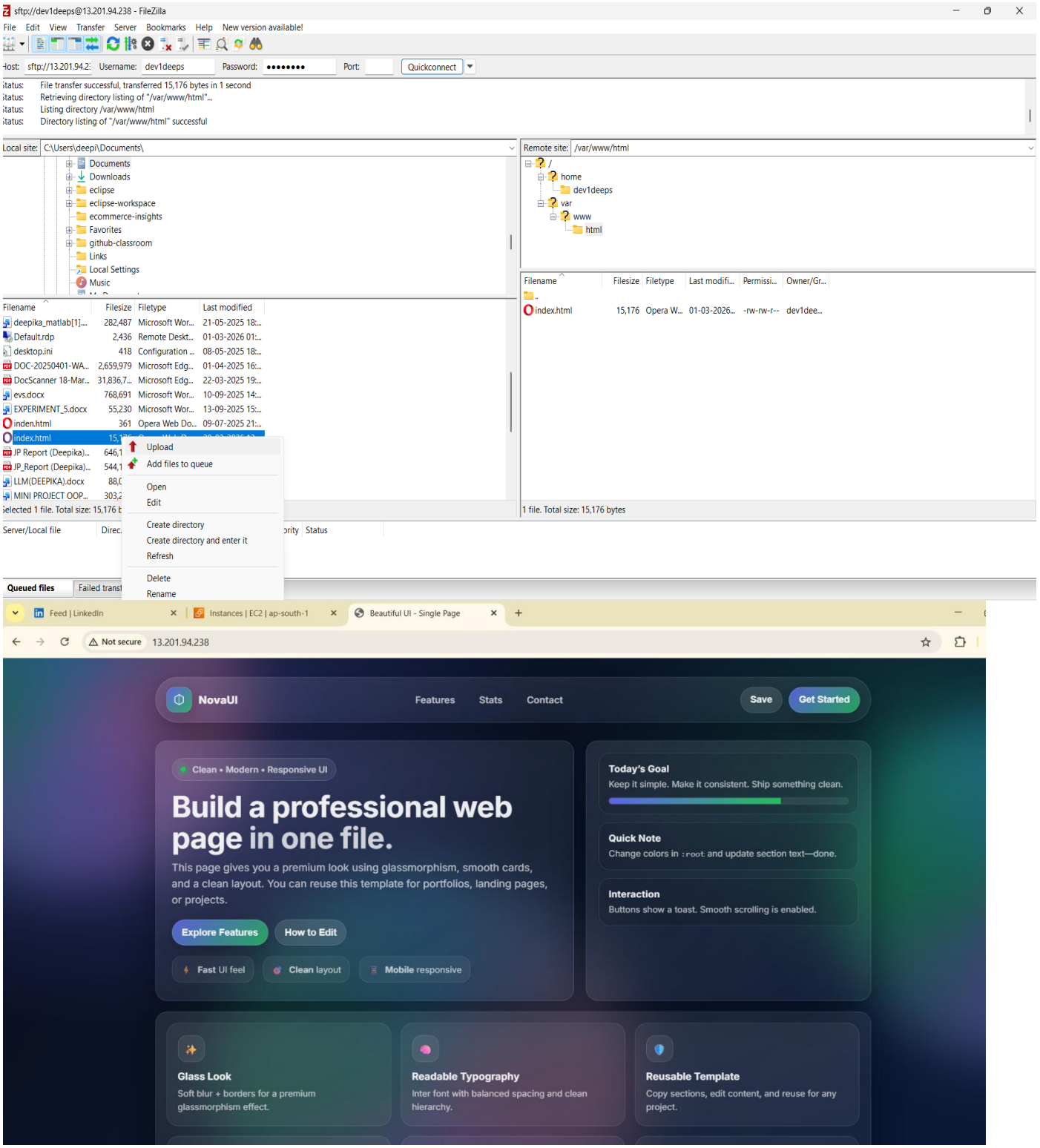




```
root@ip-172-31-28-138:~
Microsoft Windows [Version 10.0.26200.7840]
(c) Microsoft Corporation. All rights reserved.

C:\Users\deepi>cd downloads

C:\Users\deepi\Downloads>ssh -i NmindsProject1key.pem ec2-user@13.201.94.238
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Sun Mar  1 15:15:20 2026 from 103.183.240.250
[ec2-user@ip-172-31-28-138 ~]$ sudo -i
[root@ip-172-31-28-138 ~]# yum update -y
yum install hAmazon Linux 2023 Kernel Livepatch repos     [===                                  ] ---  B/s |   0  B
Amazon Linux 2023 Kernel Livepatch repository                                    221 kB/s |  31 kB     00:00
Dependencies resolved.
Nothing to do.
Complete!
[root@ip-172-31-28-138 ~]# yum install httpd -y
Last metadata expiration check: 0:00:20 ago on Sun Mar  1 15:43:13 2026.
Dependencies resolved.
================================================================================================================
 Package                        Architecture        Version                              Repository            Size
================================================================================================================
```

```
Complete!
[root@ip-172-31-28-138 ~]# systemctl start httpd
[root@ip-172-31-28-138 ~]# systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[root@ip-172-31-28-138 ~]# systemctl status httpd
● httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
     Active: active (running) since Sun 2026-03-01 15:43:50 UTC; 17s ago
       Docs: man:httpd.service(8)
   Main PID: 27067 (httpd)
     Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes served/sec:   0 B/sec"
      Tasks: 177 (limit: 1120)
     Memory: 12.9M
        CPU: 63ms
     CGroup: /system.slice/httpd.service
             ├─27067 /usr/sbin/httpd -DFOREGROUND
             ├─27069 /usr/sbin/httpd -DFOREGROUND
             ├─27070 /usr/sbin/httpd -DFOREGROUND
             ├─27071 /usr/sbin/httpd -DFOREGROUND
             └─27072 /usr/sbin/httpd -DFOREGROUND

Mar 01 15:43:50 ip-172-31-28-138.ap-south-1.compute.internal systemd[1]: Starting httpd.service - The Apache HTTP Serve
Mar 01 15:43:50 ip-172-31-28-138.ap-south-1.compute.internal systemd[1]: Started httpd.service - The Apache HTTP Server
Mar 01 15:43:50 ip-172-31-28-138.ap-south-1.compute.internal httpd[27067]: Server configured, listening on: port 80
lines 1-19/19 (END)
[root@ip-172-31-28-138 ~]# useradd dev1deeps
[root@ip-172-31-28-138 ~]# cat /etc/passwd
ec2-user:x:1000:1000:EC2 Default User:/home/ec2-user:/bin/bash
apache:x:48:48:Apache:/usr/share/httpd:/sbin/nologin
dev1deeps:x:1001:1001::/home/dev1deeps:/bin/bash
[root@ip-172-31-28-138 ~]# groupadd secedev
[root@ip-172-31-28-138 ~]# cat /etc/group
ec2-user:x:1000:
apache:x:48:
dev1deeps:x:1001:
secedev:x:1002:
[root@ip-172-31-28-138 ~]# useradd dev2
[root@ip-172-31-28-138 ~]# cat /etc/group
ec2-user:x:1000:
apache:x:48:
dev1deeps:x:1001:
secedev:x:1002:
dev2:x:1003:
[root@ip-172-31-28-138 ~]# usermod -G secedev dev1deeps
[root@ip-172-31-28-138 ~]# usermod -G secedev dev2
[root@ip-172-31-28-138 ~]# ls -ld /var/www/html
drwxr-xr-x. 2 root root 6 Dec  5 21:19 /var/www/html
[root@ip-172-31-28-138 ~]# chgrp secedev /var/www/html
[root@ip-172-31-28-138 ~]# ls -ld /var/www/html
drwxr-xr-x. 2 root secedev 6 Dec  5 21:19 /var/www/html
[root@ip-172-31-28-138 ~]# chmod 775 /var/www/html
[root@ip-172-31-28-138 ~]# ls -ld /var/www/html
drwxrwxr-x. 2 root secedev 6 Dec  5 21:19 /var/www/html
[root@ip-172-31-28-138 ~]# passwd root
Changing password for user root.
[root@ip-172-31-28-138 ~]# passwd dev1deeps
Changing password for user dev1deeps.
New password:
[root@ip-172-31-28-138 ~]# passwd dev2
Changing password for user dev2.
New password:
[root@ip-172-31-28-138 ~]# vi /etc/ssh/sshd_config
```


root@ip-172-31-28-138:~
#MaxAuthTries 6
#MaxSessions 10
PubkeyAuthentication no
The default is to check both .ssh/authorized_keys and .ssh/authorized_keys2
but this is overridden so installations will only check .ssh/authorized_keys
AuthorizedKeysFile .ssh/authorized_keys
#AuthorizedPrincipalsFile none
For this to work you will also need host keys in /etc/ssh/ssh_known_hosts
#HostbasedAuthentication no
Change to yes if you don't trust ~/.ssh/known_hosts for
HostbasedAuthentication
#IgnoreUserKnownHosts no
Don't read the user's ~/.rhosts and ~/.shosts files
#IgnoreRhosts yes
Explicitly disable PasswordAuthentication. By presetting it, we
avoid the cloud-init set_passwords module modifying sshd_config and
restarting sshd in the default instance launch configuration.
PasswordAuthentication yes
PermitEmptyPasswords no
Change to no to disable s/key passwords
#KbdInteractiveAuthentication yes
-- INSERT --
65,27
39%
sftp://dev1deeps@13.201.94.238 - FileZilla
Status: Connected to 13.201.94.238
Status: Retrieving directory listing...
Status: Listing directory /home/dev1deeps
Status: Directory listing of "/home/dev1deeps" successful
Local site: C:\Users\deepi\Documents\
Remote site: /home/dev1deeps
.bash_logout
.bash_profile
.bashrc
3 files. Total size: 651 bytes


sftp://dev1deeps@13.201.94.238 - FileZilla
File Edit View Transfer Server Bookmarks Help New version available!
Host: sftp://13.201.94.2: Username: dev1deeps Password: ••••••••• Port: Quickconnect
Status: File transfer successful, transferred 15,176 bytes in 1 second
Status: Retrieving directory listing of "/var/www/html"...
Status: Listing directory /var/www/html
Status: Directory listing of "/var/www/html" successful
Local site: C:\Users\deepi\Documents\
Remote site: /var/www/html
Upload
Add files to queue
Open
Edit
Create directory
Create directory and enter it
Refresh
Delete
Rename
1 file. Total size: 15,176 bytes
Feed | LinkedIn
Instances | EC2 | ap-south-1
Beautiful UI - Single Page
Not secure 13.201.94.238
NovaUI
Features
Stats
Contact
Save
Get Started
Clean • Modern • Responsive UI
Build a professional web page in one file.
This page gives you a premium look using glassmorphism, smooth cards, and a clean layout. You can reuse this template for portfolios, landing pages, or projects.
Explore Features
How to Edit
Fast UI feel
Clean layout
Mobile responsive
Today's Goal
Keep it simple. Make it consistent. Ship something clean.
Quick Note
Change colors in :root and update section text—done.
Interaction
Buttons show a toast. Smooth scrolling is enabled.
Glass Look
Soft blur + borders for a premium glassmorphism effect.
Readable Typography
Inter font with balanced spacing and clean hierarchy.
Reusable Template
Copy sections, edit content, and reuse for any project.